

Unit : 2
AC signals

✓ so far we have studied operⁿ of circuit ~~with~~ DC signal, now we will provide ^{for} sinusoidal signal as input.

$$V(t) = V_m \sin(\omega t)$$

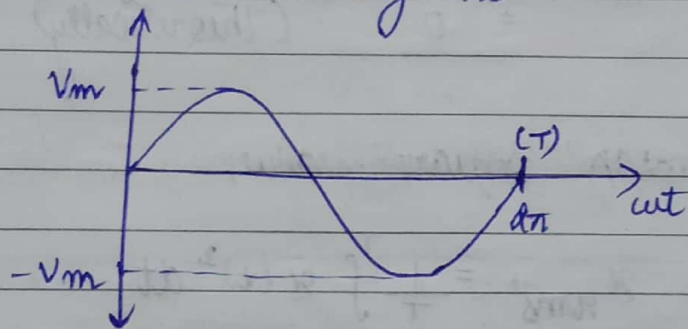
↳ nature

V_m = Amplitude

ω = frequency

t = time

ωt = argument



$$\omega T = 2\pi$$

$$\omega = \frac{2\pi}{T}$$

$$\omega = 2\pi f$$

$$2\pi \text{ (rad)} = 360^\circ$$

$$1 \text{ rad} = 57.3^\circ$$

conversion factor

$$\text{rad} = \frac{\pi}{180} \times ^\circ$$

$$^\circ = \frac{180}{\pi} \times \text{rad}$$

✓ Average value of Alternating signal:

In general, average value of general signal $x(t) = \frac{1}{T} \int_0^T x(t) dt$

$$V_{avg} = \frac{1}{\pi} \int_0^{\pi} V_m \sin d(\omega t)$$

$$= 0.63 V_m$$

for $V_{avg} = \frac{1}{\pi} \int_0^{2\pi} V_m \sin d(\omega t)$

$$= 0 \quad (\text{Theoretically})$$

Root mean square value,

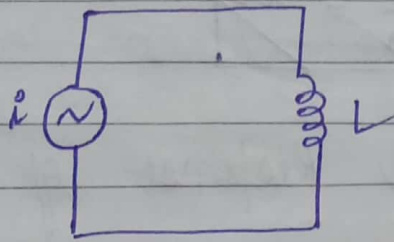
$$x_{rms} = \frac{1}{T} \int_0^T x(t)^2 dt$$

$$V_{rms} = \frac{1}{2\pi} \int_0^{2\pi} (V_m \sin \omega t)^2 d(\omega t)$$

$$= 0.707 V_m$$

✓ Response of AC signal in inductor or capacitor:

i)



(lags by 90°)

$$i = I_m \sin \omega t$$

$$V = L \frac{di}{dt}$$

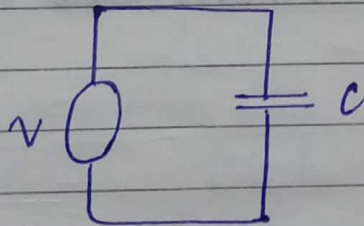
$$V = \omega L I_m \cos \omega t$$

$$= V_m \sin (\omega t + 90^\circ)$$

$$V_m = \omega L I_m$$

$$\boxed{\frac{V_m}{I_m} = \omega L}$$

ii)



(leads by 90°)

$$V = V_m \sin \omega t$$

$$i = C \frac{dV}{dt}$$

$$i = \omega C V_m \cos \omega t$$

$$i = I_m \cos \omega t$$

$$i = I_m \sin (\omega t + 90^\circ)$$

$$I_m = \omega C V_m$$

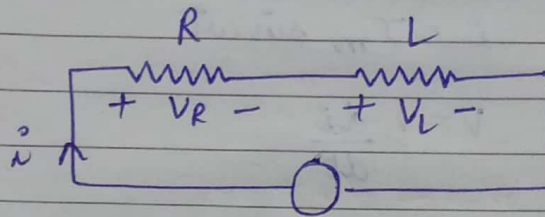
$$\boxed{\frac{1}{\omega C} = \frac{V_m}{I_m}}$$

$$\cancel{V_m = \omega L I_m}$$

$$\cancel{\frac{V_m}{I_m} = \omega L}$$

✓ #

Sinusoidal Response of series R-L circuit:



$$V_R = iR = R I_m \sin \omega t$$

$$V_L = \omega L I_m \sin(\omega t + 90^\circ)$$

$$V = V_R + V_L$$

$$V_m \sin(\omega t + \theta) = R I_m \sin \omega t + \omega L I_m \sin(\omega t + 90^\circ)$$

↑
phase shift

$$V_m \sin \omega t \cos \theta + V_m \sin \theta \cos \omega t = R I_m \sin \omega t + \omega L I_m \sin \omega t \cos 90^\circ + \omega L I_m \sin 90^\circ \cos \omega t$$

here,

$$V_m \cos \theta = R I_m$$

$$V_m \sin \theta = \omega L I_m$$

$$V_m^2 = (R I_m)^2 + (\omega L I_m)^2$$

$$V_m = I_m \sqrt{R^2 + (\omega L)^2}$$

$$V_m = I_m Z$$

↓
impedance

lag by $\frac{\omega L}{R}$

→ $R + j\omega L$

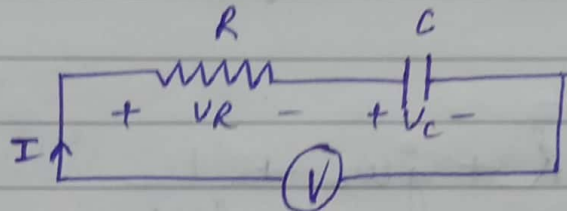
$$\sqrt{R^2 + (\omega L)^2} \angle \tan^{-1} \left(\frac{\omega L}{R} \right)$$

$$Z = \sqrt{R^2 + (\omega L)^2} \angle \tan^{-1} \left(\frac{\omega L}{R} \right)$$

$$I_m = \frac{V_m}{Z} = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \angle -\tan^{-1} \left(\frac{\omega L}{R} \right)$$

✓ #

Sinusoidal Response of series R-C circuit:



$$V = V_m \sin \omega t$$

$$V = I \left(R + \frac{1}{j\omega C} \right)$$

$$V = I \left[R - \frac{j}{\omega C} \right]$$

$$Z = R - \frac{j}{\omega C}$$

$$Z = \sqrt{R^2 + \left(\frac{1}{\omega C} \right)^2} \angle -\tan^{-1} \frac{1}{\omega R C}$$

current leads by $\frac{1}{\omega R C}$

$$I = \frac{V}{Z}$$

$$= \frac{V}{\sqrt{R^2 + \left(\frac{1}{\omega C} \right)^2} \angle -\tan^{-1} \frac{1}{\omega R C}}$$

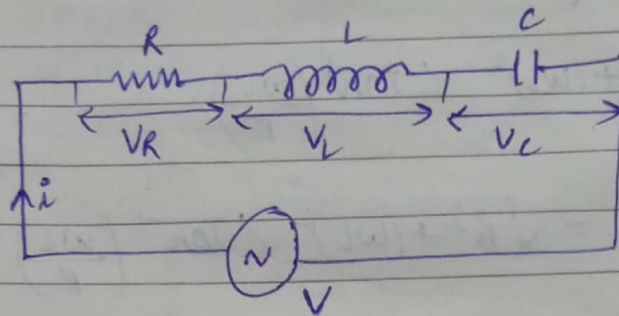
$$= \frac{V}{\sqrt{R^2 + \left(\frac{1}{\omega C} \right)^2}} \angle \tan^{-1} \left(\frac{1}{\omega R C} \right)$$

$$X_C = \frac{1}{\omega C}$$

$$= \frac{1}{0}$$

$$= \infty$$

✓ + Series RLC circuit :



$$V = V_R + V_L + V_C$$

$$= iR + iX_L + iX_C$$

$$= i \left(R + j\omega L + \frac{1}{j\omega C} \right)$$

$$\begin{cases} X_L = j\omega L = jX_L \\ X_C = \frac{1}{j\omega C} \approx \frac{-j}{\omega C} = -jX_C \end{cases}$$

$$V = IZ$$

$$Z = R + j\omega L + \frac{1}{j\omega C}$$

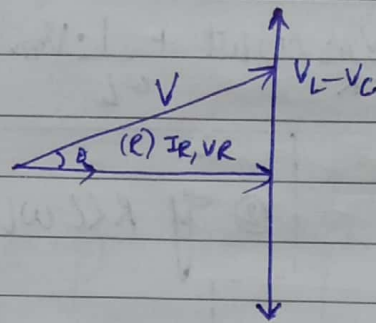
$$Z = R + j\omega L - \frac{j}{\omega C}$$

$$Z = R + j \left(\omega L - \frac{1}{\omega C} \right)$$

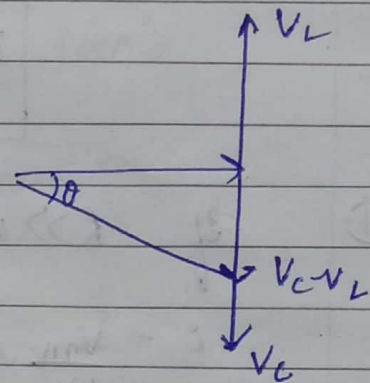
$$\text{mag.} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

$$\tan^{-1} \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right)$$

- ① circuit becomes inductive when $\omega L > \frac{1}{\omega C}$
- ② circuit becomes capacitive $\frac{1}{\omega C} > \omega L$

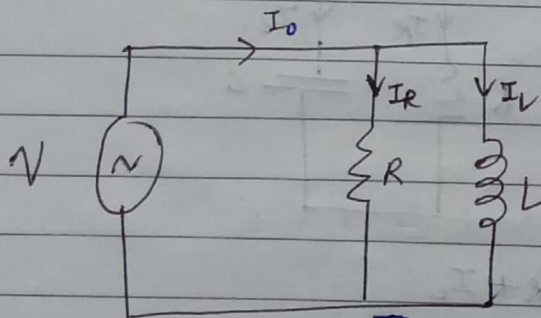


(Inductive)



(capacitive)

Sinusoidal response of RL ckt



$$V = V_m \sin \omega t$$

$$I_0 = I_L + I_R$$

$$I_R = \frac{V_m \sin \omega t}{R} = \frac{V}{R} \quad ; \quad I_L = \frac{V}{X_L}$$

$$I_0 = \frac{V}{R} + \frac{V}{X_L}$$

$$I_0 = V \left(\frac{1}{R} + \frac{1}{j\omega L} \right)$$

opposite of
impedance

γ admittance $\rightarrow$

$$\left(I = \frac{1}{Y} \right)$$

$$\gamma = \frac{1}{R} + \frac{1}{j\omega L} = \frac{1}{R} - \frac{j}{\omega L}$$

Sinusoidal i/p, $v = v_m \cos \omega t$

$$I = I_R + I_L$$

$$I = \frac{v}{R} + \frac{1}{L} \int v dt$$

$$I = \frac{V_m \cos \omega t}{R} + \frac{1}{\omega L} V_m \sin \omega t$$

① if $R \gg \omega L$

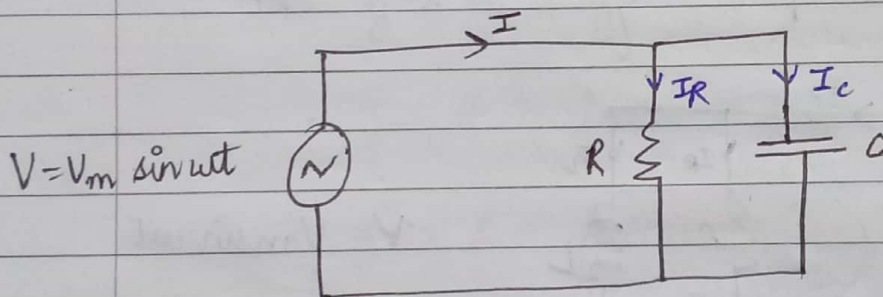
② if $R \ll \omega L$

Phase

$$i = \frac{V_m}{\omega L} \cos (\omega t - 90^\circ)$$

$$i = \frac{V_m}{R} \cos \omega t$$

Sinusoidal series of ||^l R-L circuit.



$$I = I_R + I_C$$

$$I = \frac{v}{R} + \frac{v}{\cancel{L} \omega C j \omega C}$$

$$I = \frac{v}{R} + j v \omega C$$

$$I = I_R + I_C$$

$$= \frac{V_m \sin \omega t}{R} + C \frac{dv}{dt}$$

$$= \frac{V_m \sin \omega t}{R} + \omega C V_m \cos \omega t$$

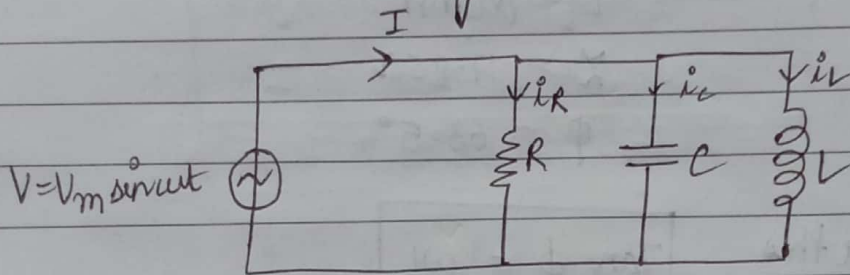
i) if $R \gg X_C$

$$i = i_C = \omega C V_m \sin(\omega t + 90^\circ)$$

ii) if $R \ll X_C$

$$i \approx i_R \approx \frac{V_m}{R} \sin \omega t$$

Sinusoidal series of RLC circuit



$$I = i_R + i_C + i_L$$

$$I = \frac{V}{R} + \frac{V}{X_L} + \frac{V}{X_C}$$

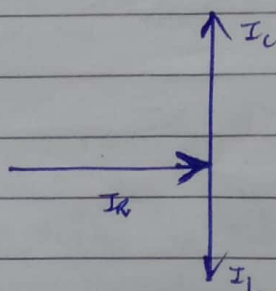
$$I = \frac{V_m}{R} \sin \omega t$$

$$I = \frac{V}{R} + \frac{V}{j\omega L} + \frac{V}{\frac{1}{j\omega C}}$$

$$I = \frac{V}{R} + \frac{V}{j\omega L} + Vj\omega C$$

$$I = \frac{V}{R} + \frac{1}{L} \int V dt + C \frac{dV}{dt}$$

$$= \frac{V_m \sin \omega t}{R} - \frac{V_m \cos \omega t}{\omega L} + V_m \omega C \cos \omega t$$



(all values in rms)

- Q In a R-L circuit the inductance being 20 millihenry (mH), impedance 17.85Ω ϕ angle of the lag of the input current from the applied voltage. find the value of R & L .

$$\omega = 796.875 \text{ rad/s}$$

$$R = 8 \Omega$$

$$L = 20 \text{ mH}$$

$$Z = 17.85 \Omega$$

$$\phi = 63.5^\circ$$

Use this

$$\tan \phi = \frac{\omega L}{R}$$

- Q Applied voltage is 11^L RLC circuit is given by,

$$v = 50 \sin (5000t + \frac{\pi}{4}) \text{ V}$$

$$\& R = 20 \Omega$$

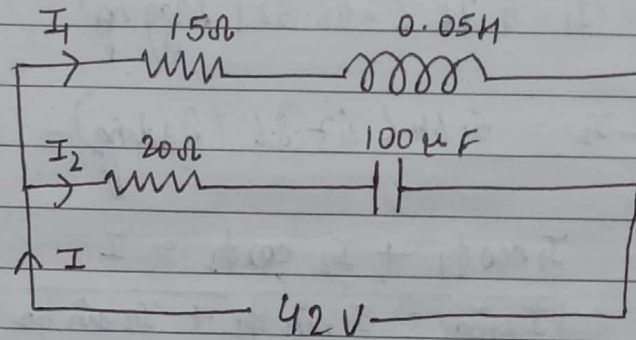
$$L = 1.6 \times 10^{-3} \text{ H}$$

$$C = 20 \mu \text{ F}$$

find total current?

$$I = 1.975 \angle 18.5^\circ \text{ A}$$

Q Determine rms value of current in each branch and total current of the circuit draw its phasor diagram.



$$f = 50 \text{ Hz}$$

$$Z_1 = R_1 + j\omega L_1$$

$$= 15 + j(2\pi) \times 50 \times 0.05$$

$$= 15 + j(15.71)$$

$$Z_1 = \sqrt{(15)^2 + (15.71)^2}$$

$$= 21.72 \Omega$$

$$\tan^{-1} \left(\frac{15.71}{15} \right) = \angle 46.32^\circ \text{ (lagging voltage)}$$

$$Z_2 = 20 + (-j) \frac{1 \times 10^4}{2\pi \times 50 \times 100 \times 10^{-6}}$$

$$= 20 + (-j) \times \frac{100}{\pi}$$

$$= 20 - 31.81j$$

$$|Z_2| = 37.58$$

$$\angle 57.86^\circ \text{ (leading)}$$

$$I_1 = \frac{V}{Z_1} = \frac{212}{21.72 \angle 46.32}$$

$$I_2 = \frac{212}{37.58}$$

$$\angle 57.86$$

$$I_1 = 9.76 \angle -46.32 \text{ (lagging)}$$

$$I_2 = 5.64 \angle 57.86 \text{ (leading)}$$

$$I_1 \cos \phi_1 + I_2 \cos \phi_2 = I$$

$$I_{\text{img}} = I_1 \sin \phi_1 + I_2 \sin \phi_2$$

$$I = I_R + j I_{\text{img}}$$

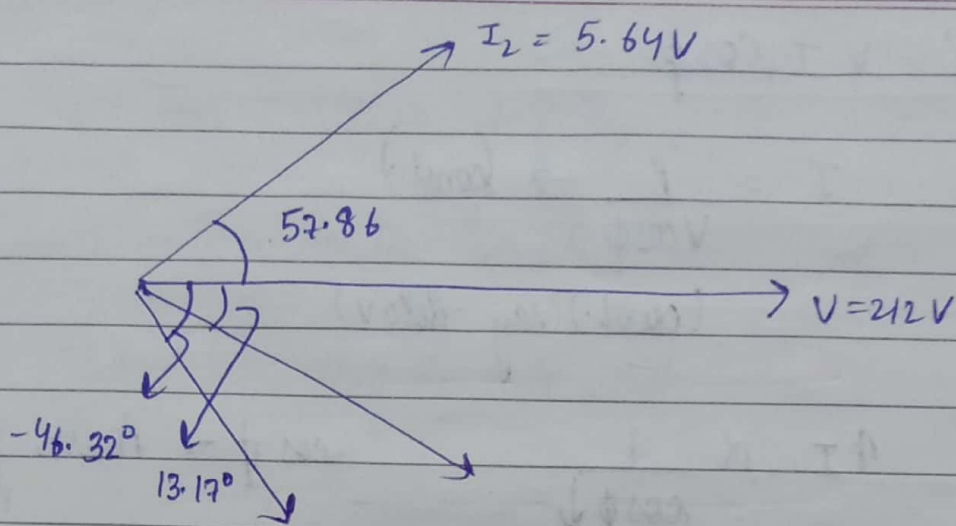
$$\phi = \tan^{-1} \left(\frac{I_I}{I_R} \right)$$

$$I_{\text{Real}} = 9.76 \cos(46.32) + 5.64 \cos(57.86) = 9.74$$

$$I_{\text{img}} = -2.28$$

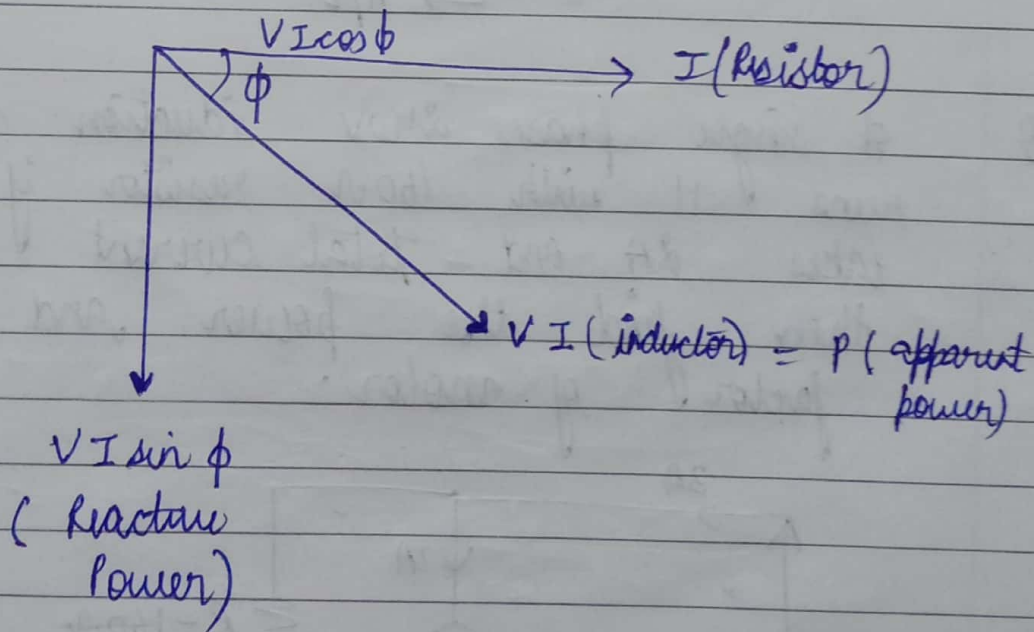
$$\phi = \tan^{-1} \left(\frac{I_{\text{img}}}{I_R} \right) = -13.17^\circ$$

Reference line (which is common) in this case voltage is common.



$$\frac{1}{X_L} = \text{susceptance}$$

✓ Conductance + susceptance = admittance (Ω^{-1})
 ($X_C \propto X_L$)⁻¹
 Real / actual / active power



Resistor also has $P = VI \cos \phi$
 But ϕ is 0

Real power $\rightarrow$ power consumed by load
 Reactive power $\rightarrow$ not consumed by load.

$$P = VI \cos \phi$$

$$I = \frac{P}{V \cos \phi} \rightarrow (\text{const.})$$

(const.) (say 240V)

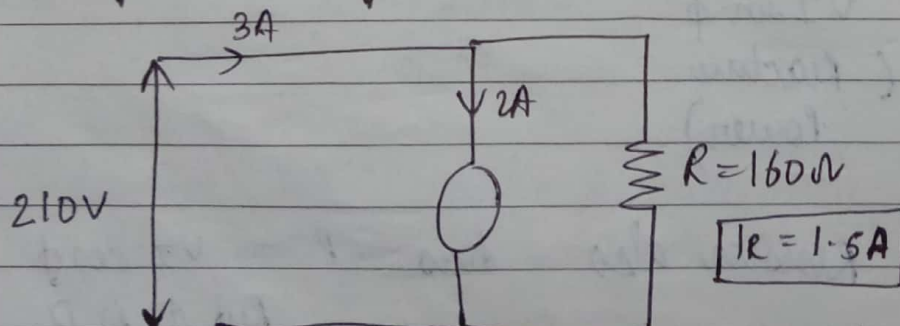
$$\uparrow I \propto \frac{1}{\cos \phi \downarrow} \quad \cos \phi \rightarrow \text{Power factor}$$

if $\cos \phi < 1$, i increases (i.e. equip is drawing more current from source)

Real power consumed by inductor/capacitor = 0

$$VI \cos \phi \rightarrow \pi/2$$

Q A single phase 240V induction motor runs || with 160Ω resistor if motor takes 2A and total current is 3A, then find the power and power factor of motor.



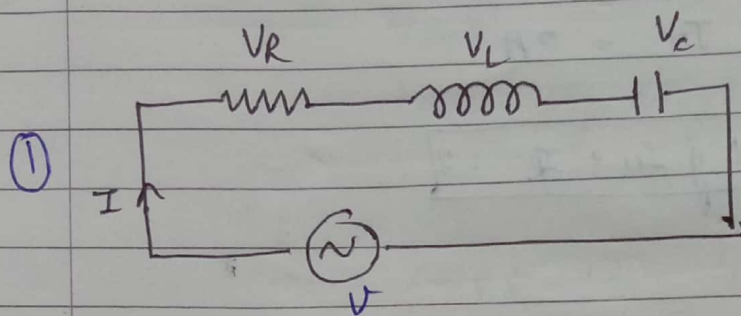
there will be lagging power factor in motor since more current is drawn than should be

$$I_m = I_{wt} + j I_{pr}$$

$$I_m + I_r = 3A$$

$$I_w + j I_u + I_r = 3$$

Series Resonance:



$$I = \frac{V}{Z}$$

$$Z = R + j\omega L + \frac{1}{j\omega C}$$

$$Z = R + j(X_L - X_C)$$

$$I = \frac{V}{R + j(X_L - X_C)}$$

at resonance

$$X_L = X_C$$

$$I = \frac{V}{R}$$

② $\cos \phi = \frac{R}{Z}$

$$X_L = X_C$$

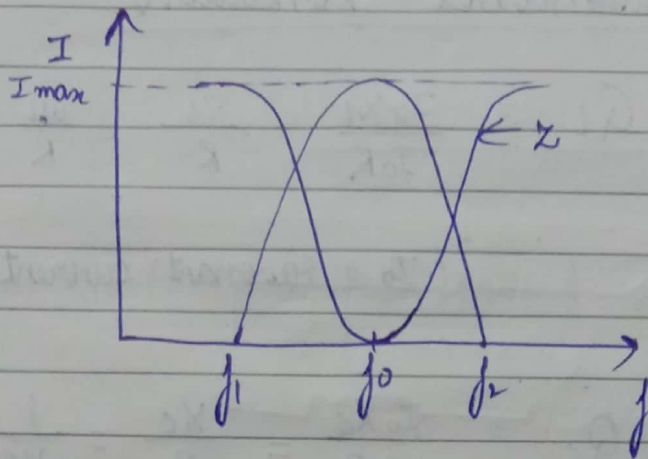
$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 = \frac{1}{LC}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

③

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$



f_1 & f_2 : Half power frequency
 Lower half power $\rightarrow$ f_1
 Higher half power $\rightarrow$ f_2

Q factor at resonance:

$$Q = \frac{V_L}{V} = \frac{V_C}{V}$$

When not
in resonance
condition

$$Q_L = \frac{I \times X_L}{I \times R} = \frac{I \times X_L}{I \times R} = \frac{I_L X_L}{I R}$$

$$Q_C = \frac{I_C \times X_C}{I \times R} = \frac{I_C X_C}{I R}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 R C}$$

In resonance condition:

$$Q_L = \frac{I_0 X_L}{I_0 R} = \frac{X_L}{R} = \frac{\omega L}{R} = \frac{1}{\sqrt{LC}} \cdot \frac{L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

(I_0 = resonant current)

$$Q_C = \frac{I_0 X_C}{I_0 R} = \frac{X_C}{R} = \frac{1}{\omega_0 R C} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

(I_0 = resonant current)

✓ Bandwidth of series resonating ckt →

at resonance : $X_L = X_C$

at half power » :

$$X = \pm (X_L - X_C) = R$$

higher ω X_L exceed X_C

lower " X_C " X_L

for f_2 ,

$$\left(\omega_2 L - \frac{1}{\omega_2 C} \right) = R \quad \text{--- (1)}$$

for f_1 ,

$$\left(\omega_1 L - \frac{1}{\omega_1 C} \right) = -R \quad \text{--- (2)}$$

(Book में काफी है पर जितना class में
कराया है इतना ही करना)

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Adding ① & ②,

$$(\omega_2 + \omega_1)L - \frac{1}{C} \left(\frac{\omega_1 + \omega_2}{\omega_1 \omega_2} \right) = 0$$

$$L = \frac{1}{C} \left(\frac{1}{\omega_1 \omega_2} \right)$$

$$\boxed{\omega_1 \omega_2 = \frac{1}{LC}} \quad \text{--- ③}$$

Subtracting ① & ② and divide by L:

then we'll get

$$\boxed{(\omega_2 - \omega_1) = \frac{R}{L}} \quad \text{--- ④}$$

$$Q = \frac{\omega_0 L}{R}$$

$$\boxed{\frac{R}{L} = \frac{\omega_0}{Q}} \quad \text{--- ⑤}$$

from eqⁿ --- ④

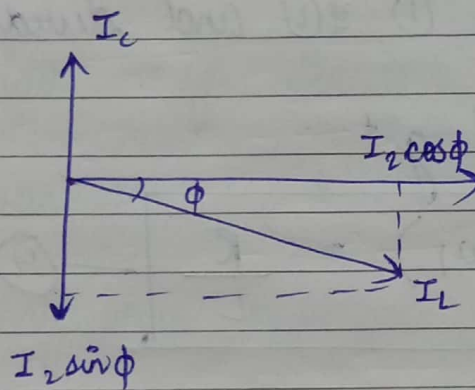
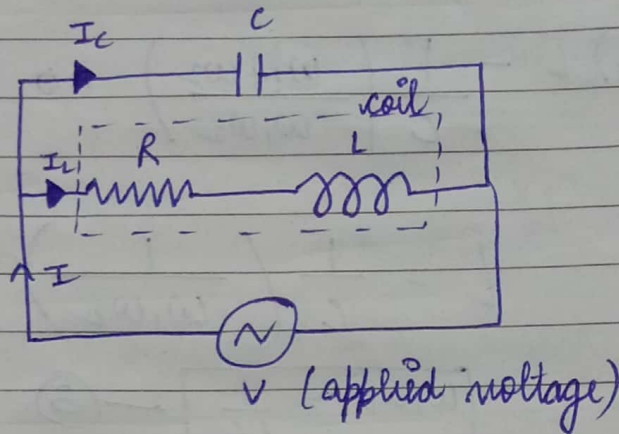
$$\omega_2 - \omega_1 = \frac{\omega_0}{Q}$$

$$\boxed{Q = \frac{\omega_0}{\omega_2 - \omega_1}} \quad = \frac{f_0}{f_2 - f_1} \quad = \frac{\text{resonant } \omega}{\text{Bandwidth}}$$

Imp. 0 Derivation of calculating Bandwidth

Date

Parallel resonance



$$I = I_L \cos \phi$$

★ Resonance, जहाँ आवृत्ति का imaginary part zero होता है

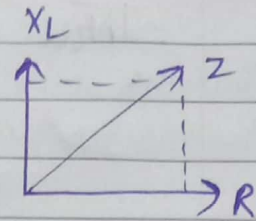
∴ at resonance,

$$I_C = I_L \sin \phi$$

$$\frac{V}{X_C} = \frac{V}{(R + jX_L)} \times \frac{X_L}{Z_L}$$

$$R + jX_L = Z_L$$

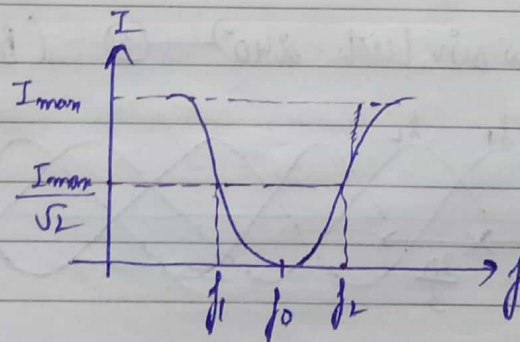
$$\frac{V}{X_C} = \frac{V}{Z_L} \times \frac{X_L}{Z_L}$$



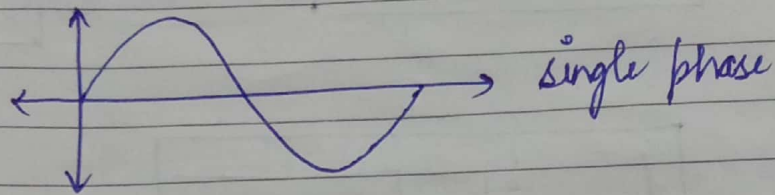
$$f_0 = \frac{1}{2\pi L} \sqrt{\frac{L}{C} - R^2}$$

$$\frac{V}{Z_{\Omega}} = \frac{V}{Z_L} \times \frac{R}{Z_L}$$

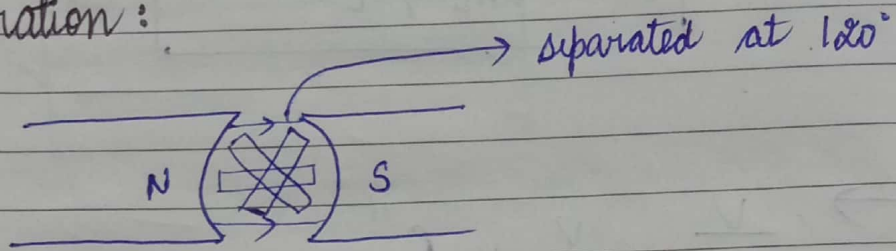
$$Z_{\Omega} = \frac{L}{Ck}$$



Three phase signals (for uninterrupted power supply)



→ Generation:

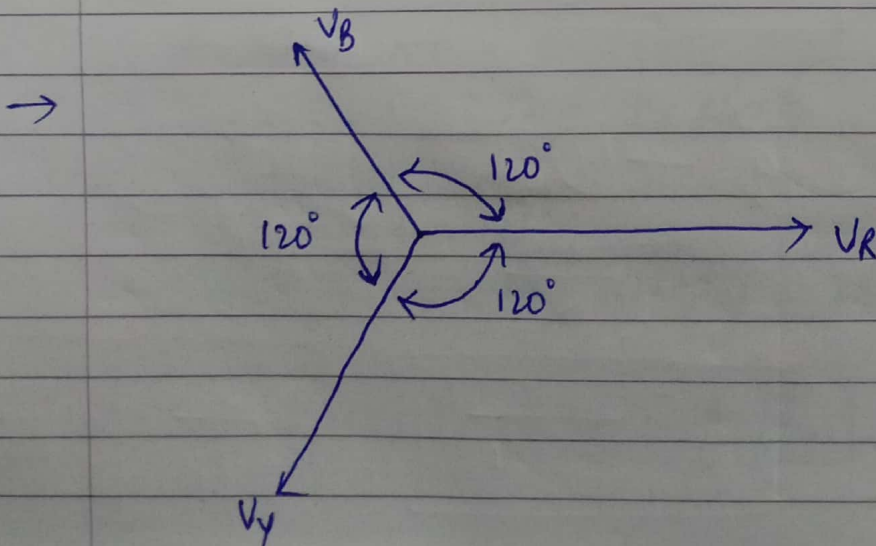
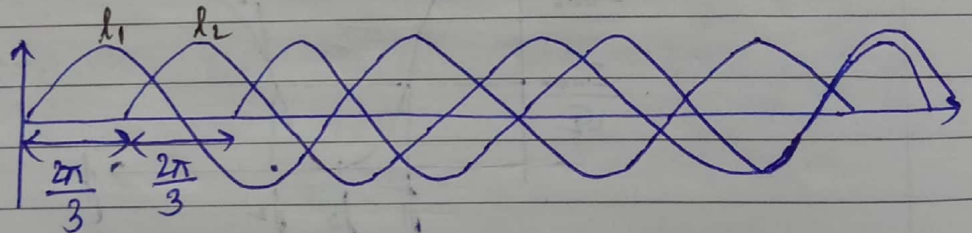


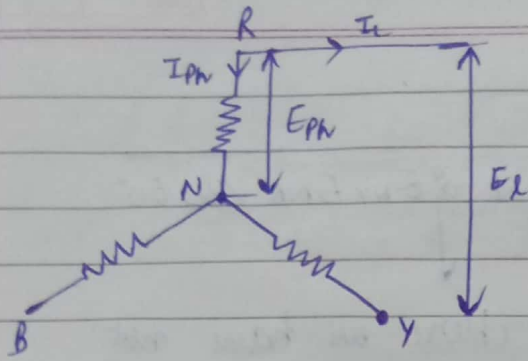
$$e = - \frac{d\phi}{dt}$$

$$e_1 = E_m \sin \omega t \text{ — ① (Red colour)}$$

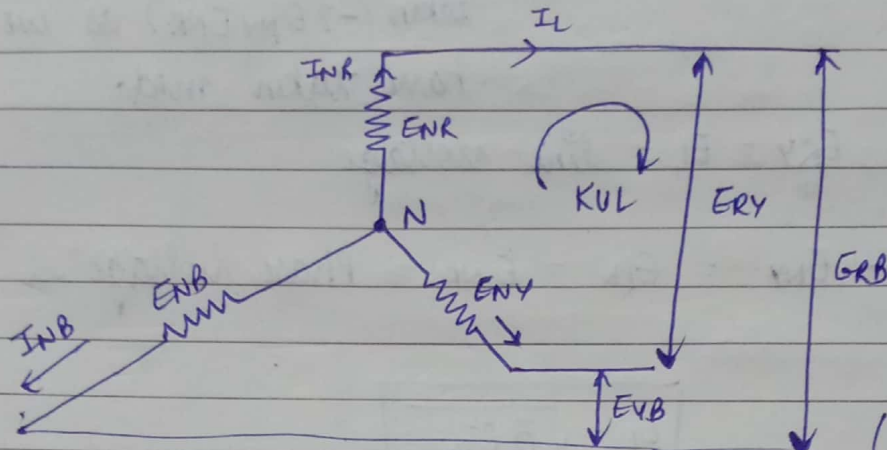
$$e_2 = E_m \sin(\omega t - 120^\circ) \text{ — ② (Yellow colour)}$$

$$e_3 = E_m \sin(\omega t - 240^\circ) \text{ — ③ (Blue colour)}$$





$I_L = \text{line current} = I_{NR}$
 $\Rightarrow \text{phase current}$

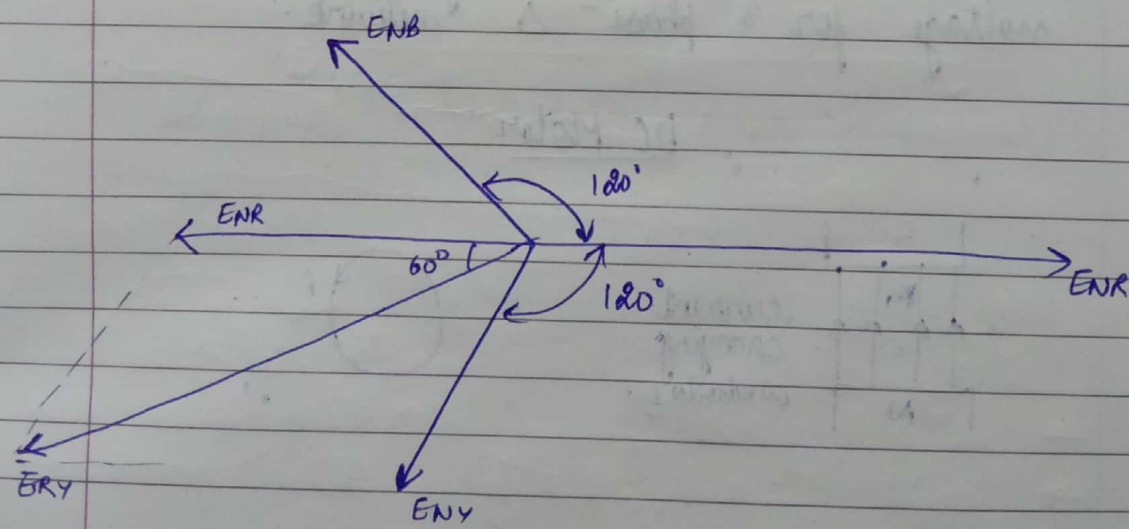


(all parts are identical)

$$\vec{E}_{NR} + \vec{E}_{RY} - \vec{E}_{NY} = 0$$

$$\vec{E}_{NR} + \vec{E}_{RY} = \vec{E}_{NY}$$

$$\vec{E}_{RY} = \vec{E}_{NY} - \vec{E}_{NR}$$



By Ilgm law:

$$E_{RY} = \sqrt{E_{NY}^2 + E_{NR}^2 + 2E_{NY}E_{NR}\cos 60^\circ}$$

(Here we have not taken $(-2E_{NY}E_{NR})$ as we have taken med.

$$E_{RY} = E_L = \text{line voltage}$$

if given points are identical $\leftarrow E_{NY} = E_{Ph} = E_{NR} = \text{Phase voltage}$

$$E_L = \sqrt{3} E_{Ph}$$

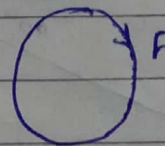
$$E_L = \sqrt{3} E_{Ph}$$

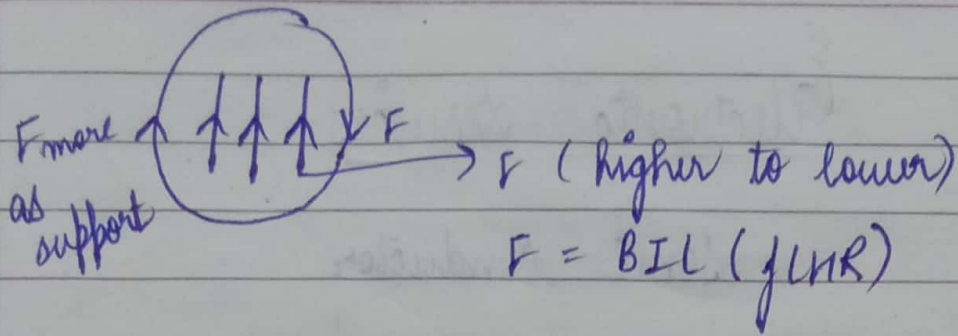
Q find relationship ① line current and phase current, ② line voltage and phase voltage for 3 phase Δ network.

DC Motor



current carrying conductor





Back emf

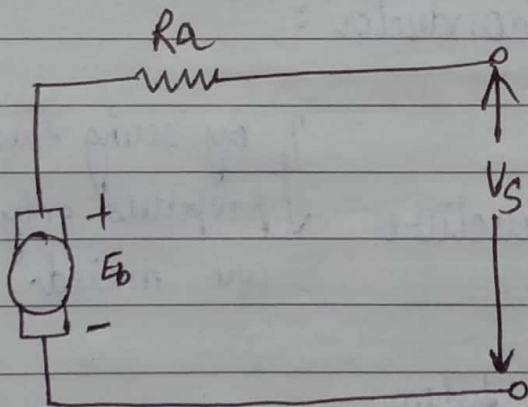
$$E_b = \frac{\phi P N Z}{60 A}$$

ϕ = flux

P = no. of poles

N = speed of rotation of conductor

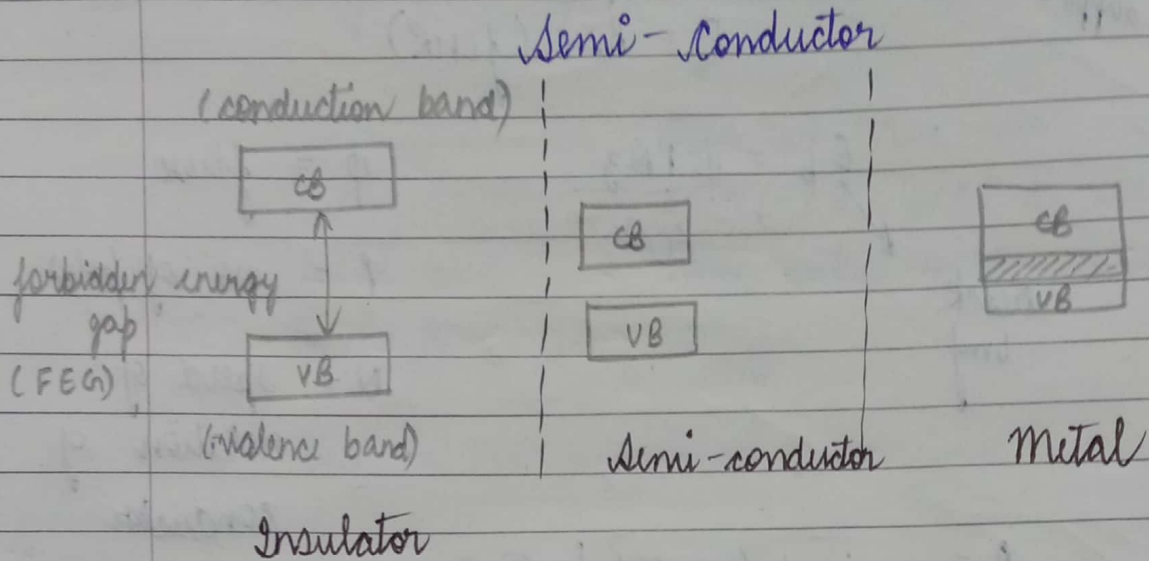
A = no. of parallel paths



(I_a = armature current)

$$V_s = E_b + I_a R_a + V_{\text{brush}}$$

Electronic Devices



Compound semiconductor:

→ Prop

→ Atomic Structure

By seeing these 2 properties only semi-conductors are mixed.

Types

↓
Simple

↓
Compound

Why Si is preferred over Ge?

Q

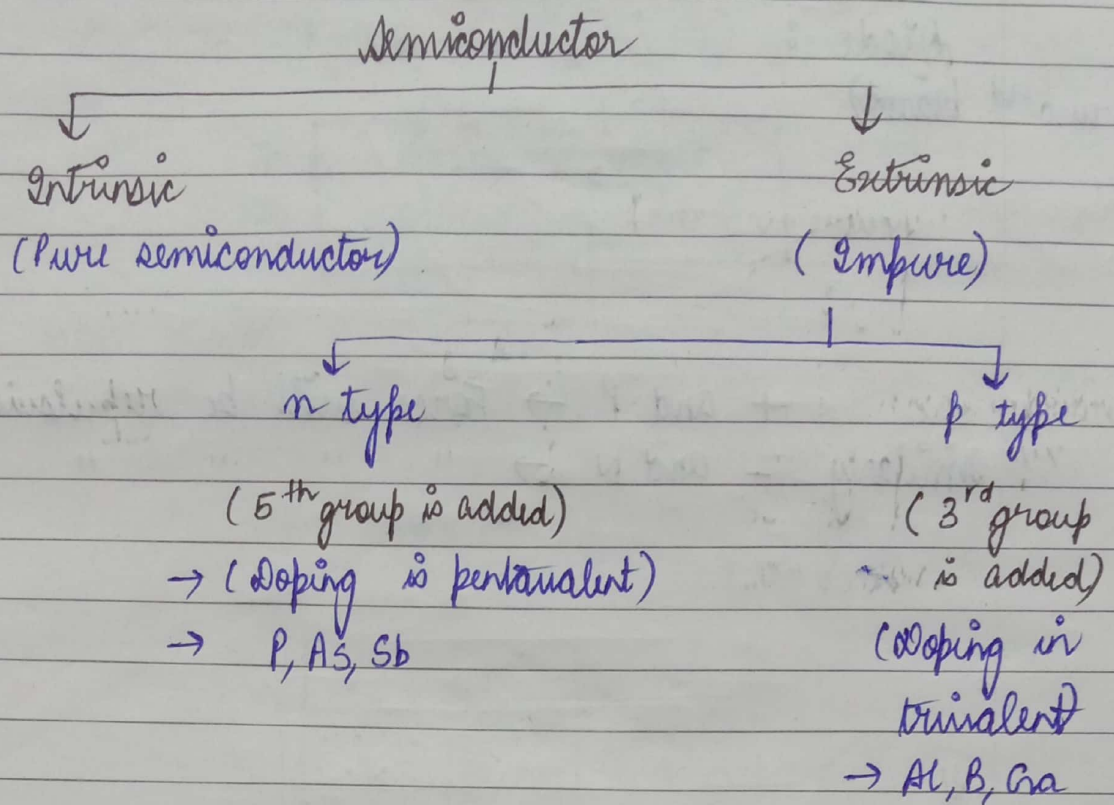
Si:

→ Easily available

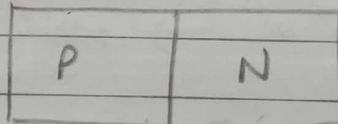
→ Temp^r withstand capability

Germanium:

→ FEG is less than Si in Germanium.



Diffusion current:



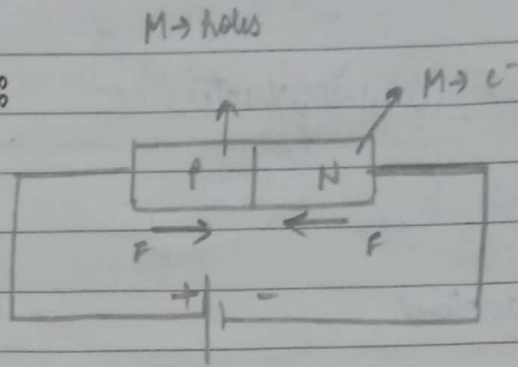
→ higher concⁿ to lower concentraⁿ
flow of e⁻ will result in Diffusion current

Drift current:

→ If voltage is applied across P & N then, the current is flowed, this results in drift current.

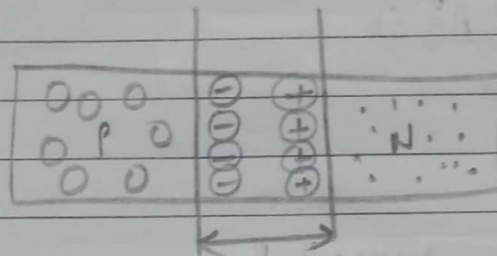
Reverse saturation current:

Diode :
(forward biased)



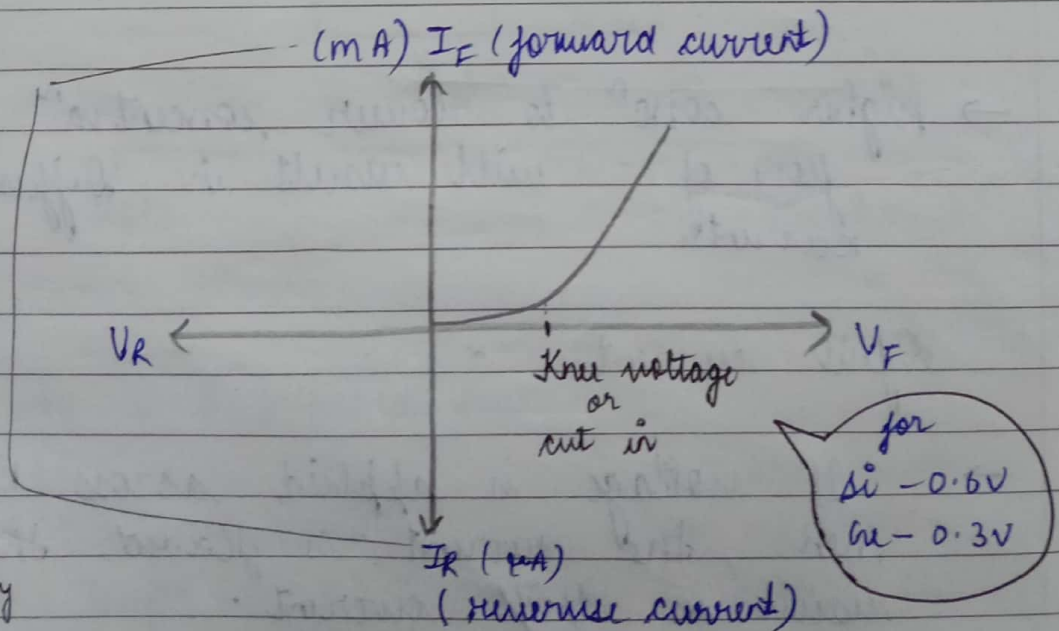
+ and P $\rightarrow$ Force will be repulsive
Similarly - and N $\rightarrow$ " " " "

If $V > V_0$
then only
current
will flow



$\rightarrow$ Depletion layer
 $\rightarrow$ Barrier potential V_0

$\rightarrow$ unit
change is
there,
because it
depends
upon
forbidden energy



$$I_D = I_0 (e^{V/nV_T} - 1)$$

I_D = Diode current

I_0 = saturation current or reverse saturation current

material $\rightarrow n = 1$ (Si)

$n = 2$ (Ge)

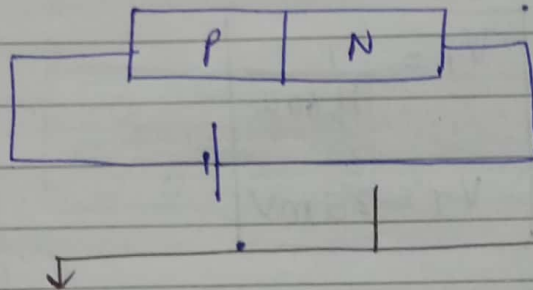
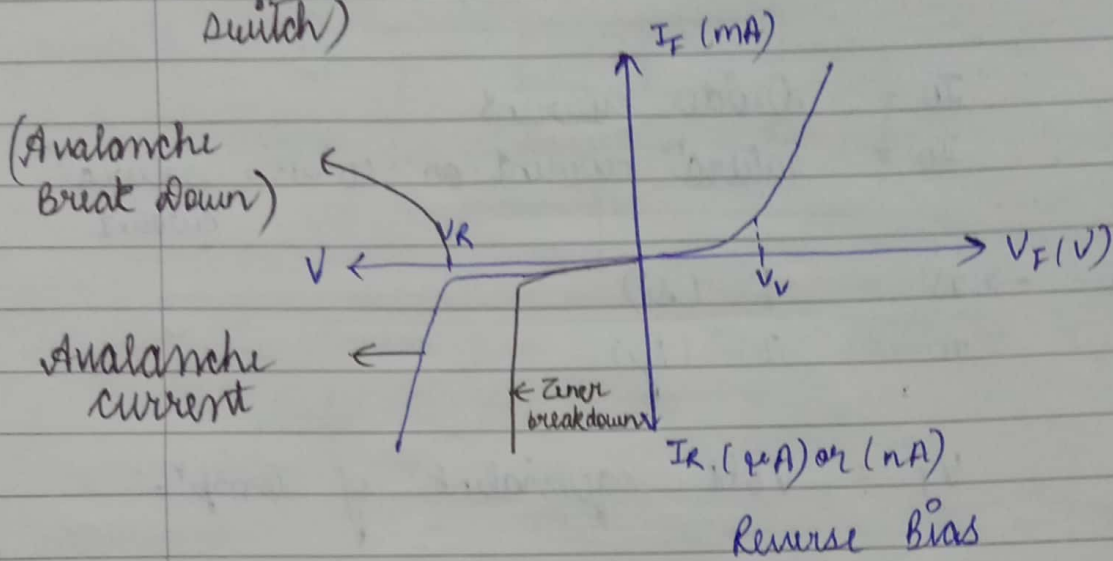
V_T = Volt equivalent of temp^r.

$V_T = \frac{T}{11,600}$
$V_T = 25 \text{ mV}$

$\rightarrow$ this device is exponential in nature.

(breakdown term is for reverse bias)

Diode characteristics :
(acts a switch)



① Zener Breakdown
(doping level is very high)

② Avalanche Breakdown

avalanche multiplication of current :
due to increase in KE of carriers.

Zener diode

$$E_d = \frac{V}{d}$$

(Reverse saturation)
(Barrier potential)

→ Effect of temp^r

Application of diode

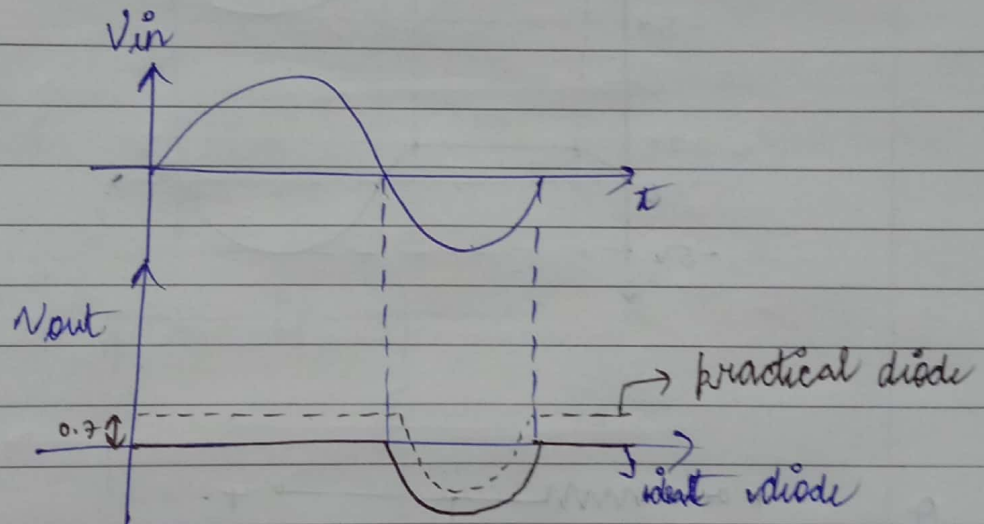
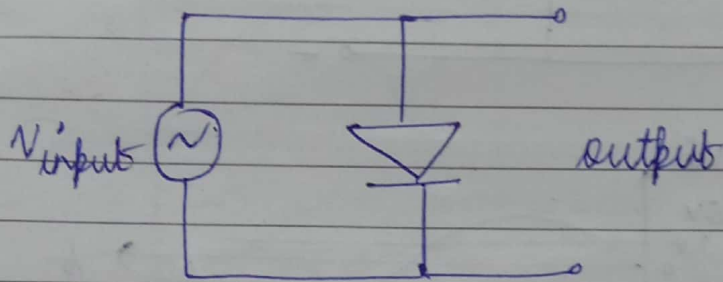
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Clipper :

→ series clipper (diode series with input supply)

→ Parallel clipper (diode in $||^e$ with input supply)

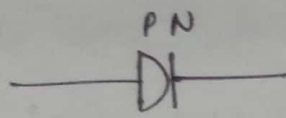
Parallel



(considering diode as ideal diode for practical diode is 0.7 drop)

forward bias
 $I = \infty$
 $V = 0$

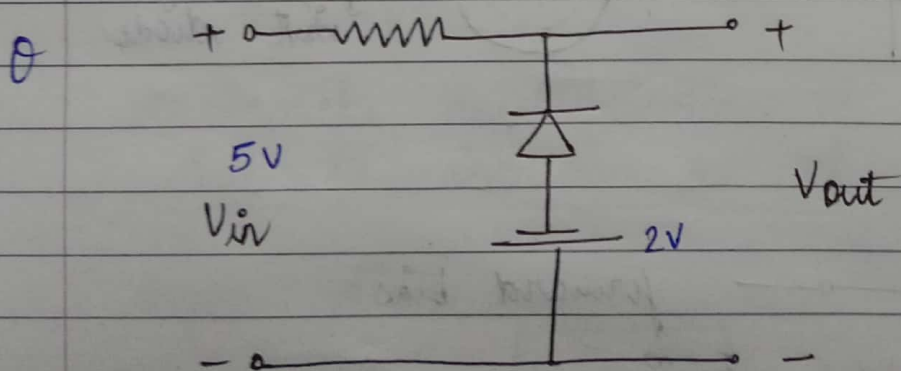
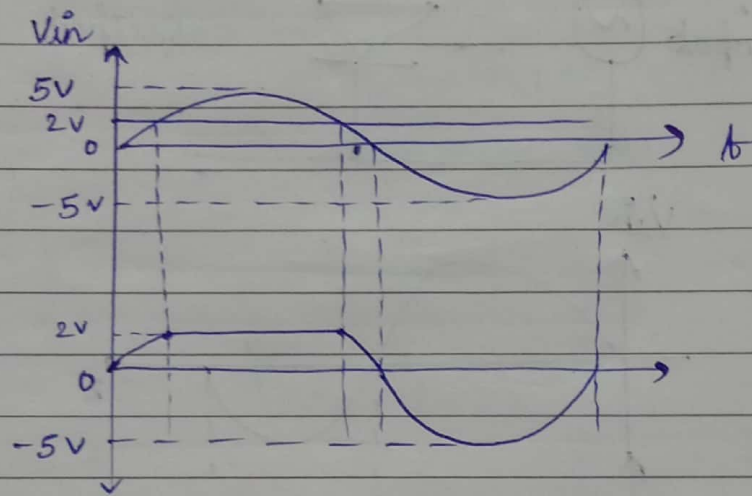
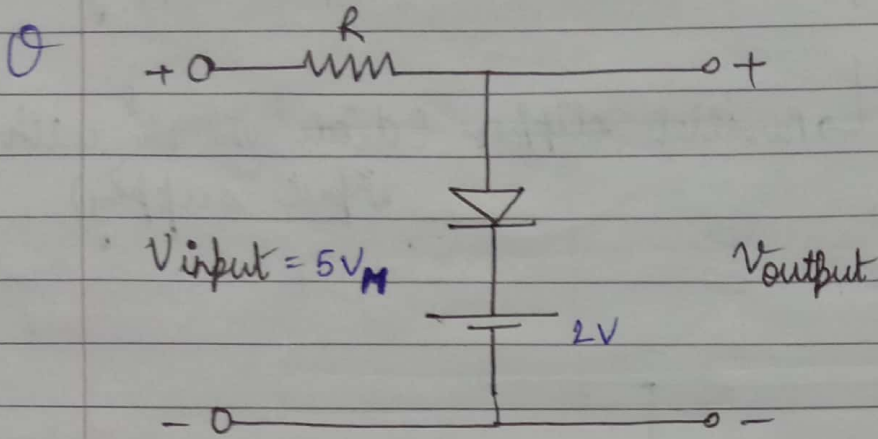
reverse bias
 $I = 0$
 $V = \infty$

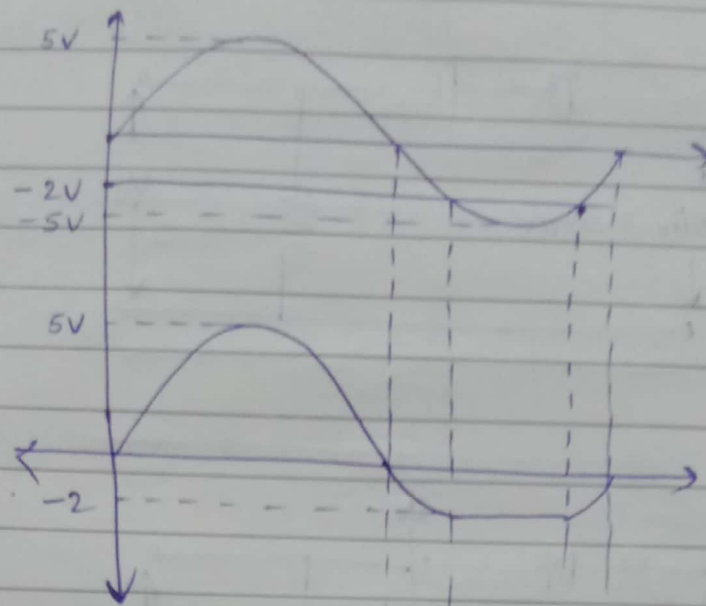


$P > N$ $\Rightarrow$ conduct

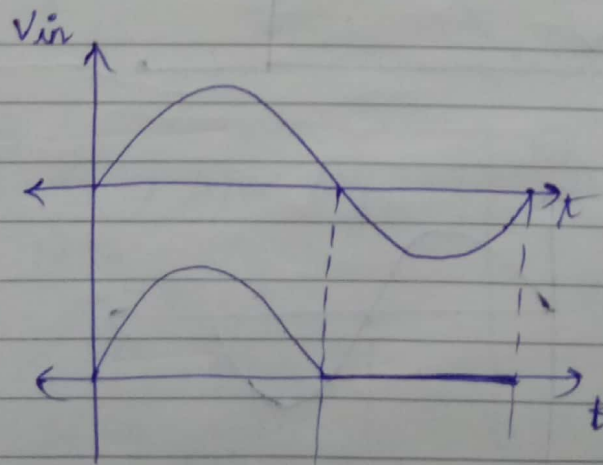
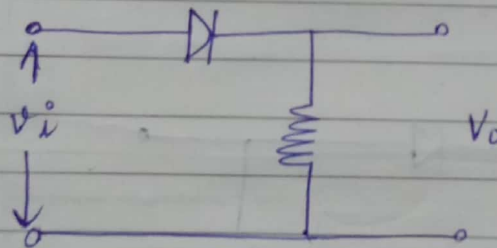
$N > P$ $\Rightarrow$ not conduct

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Date	





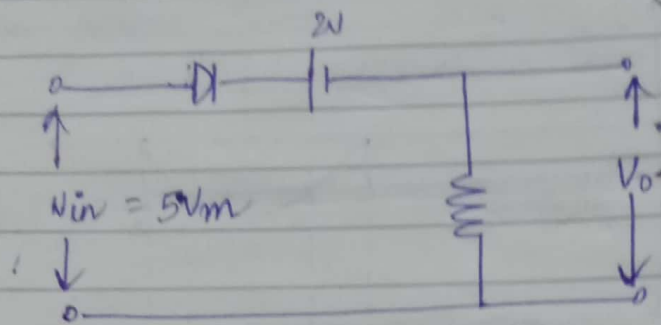
Series clipper



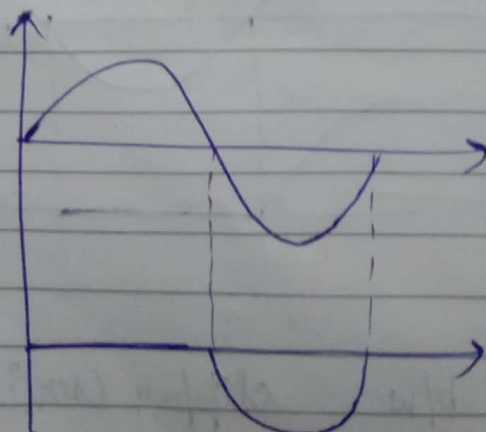
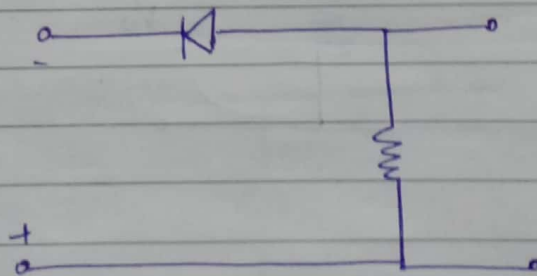
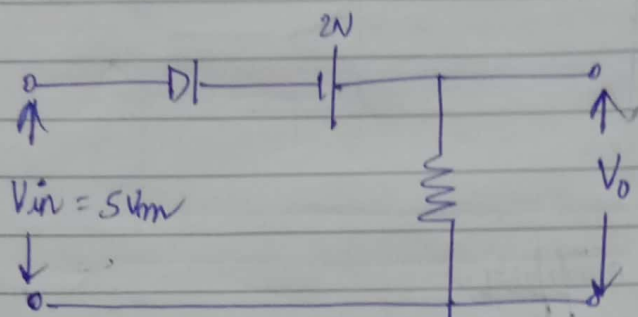
* Q Diff b/w clipper (series) & rectifier.

→ Transfer characteristics.

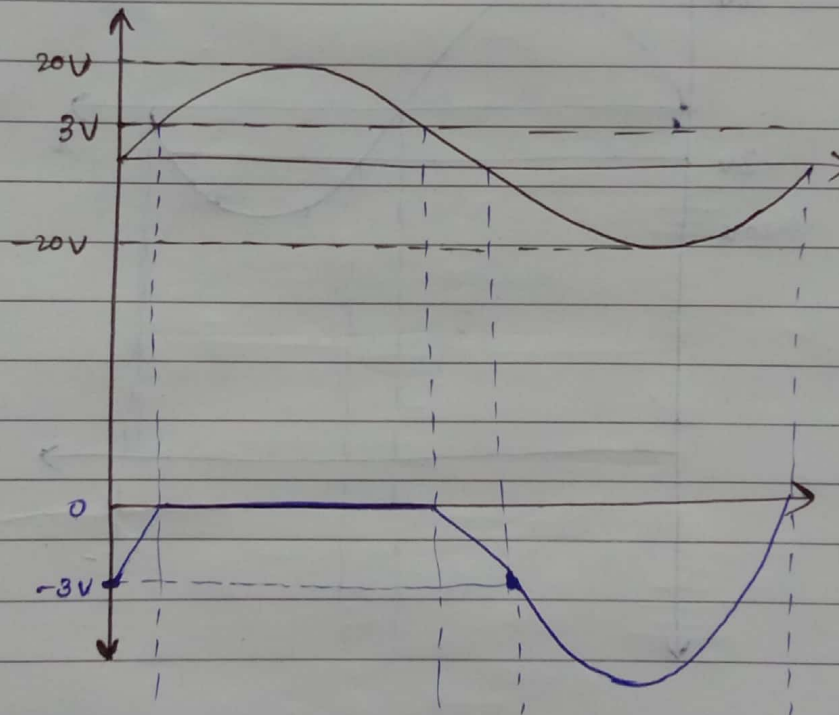
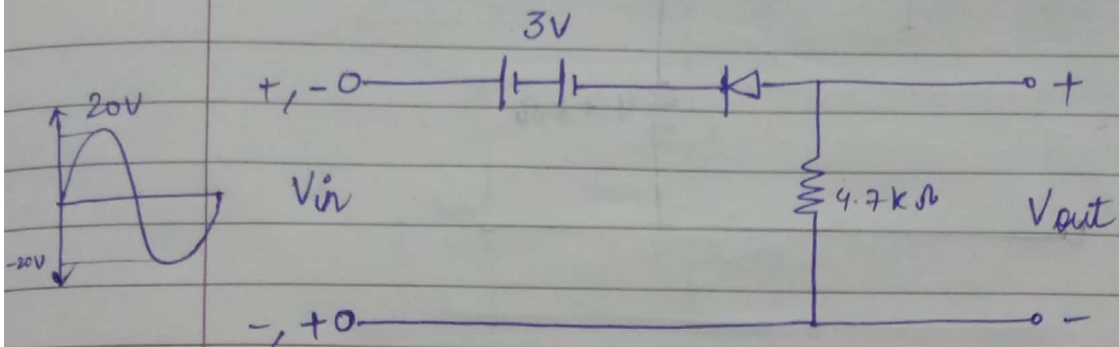
Q



Q



Series clipper:



① $V_i \rightarrow (+)ve$ half

FB $\rightarrow$

$$V_i < 3$$

$$V_o = V_i - 3V$$

RB $\rightarrow$

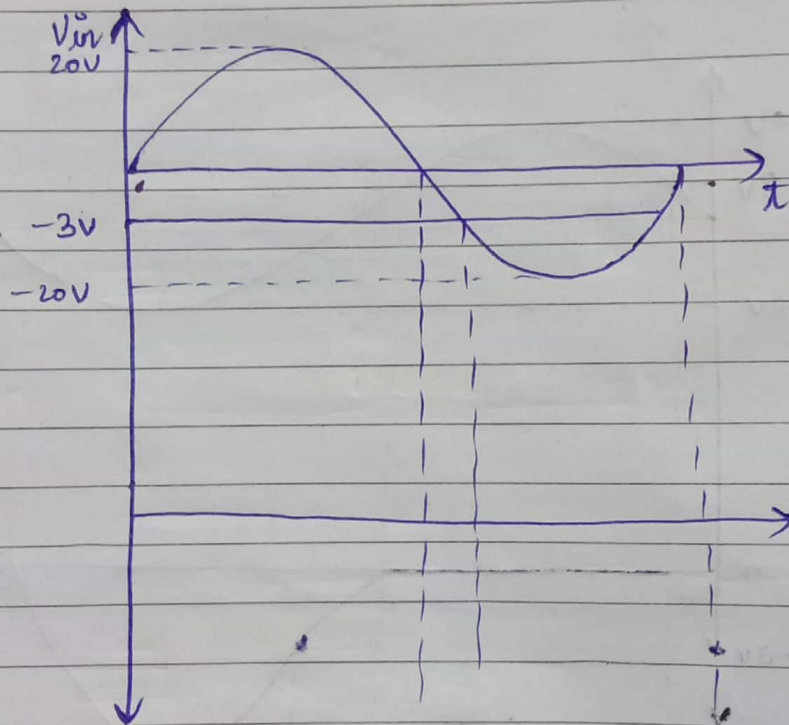
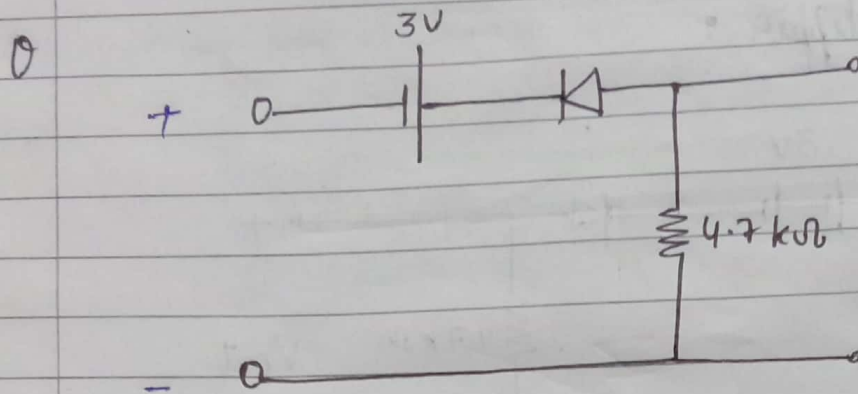
$$V_i > 3V$$

$$V_o = 0V$$

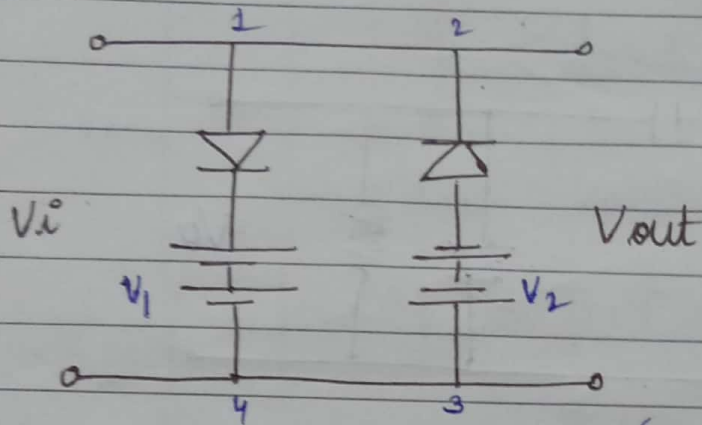
②

$V_i \rightarrow (-)ve$ half

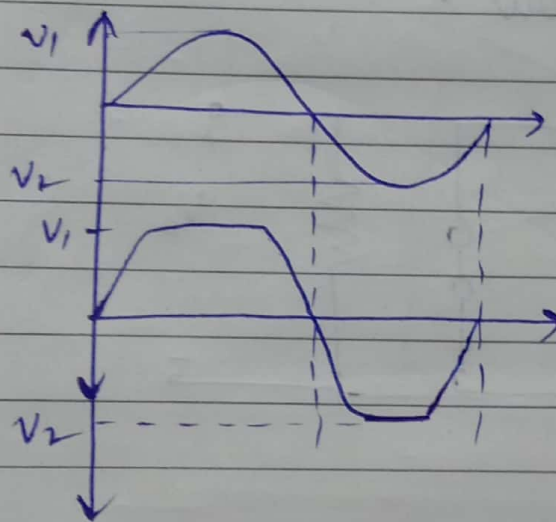
$$V_o = -V_i - 3V$$



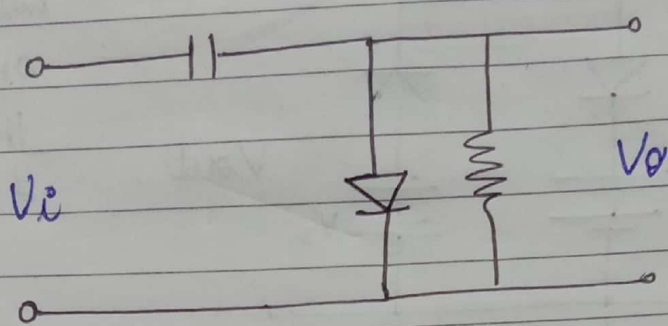
Series clipper :
 or licer or biased clipper



1-4 segment
 will work
 only when
 there is (+)ve inputs
 & also $V_2 > V_1$

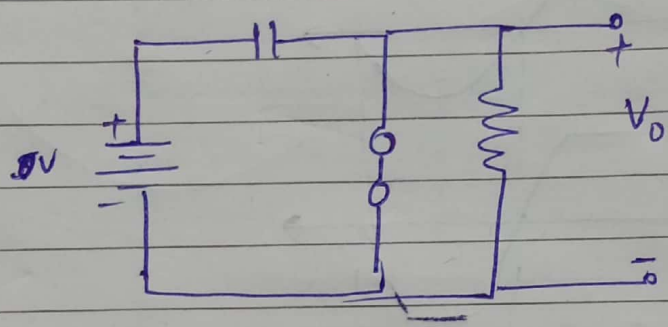


clamper circuit



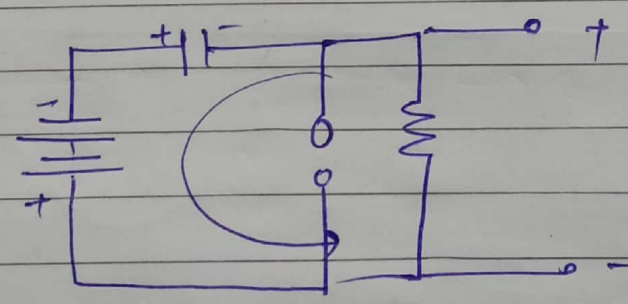
Rules to follow:

1.



when diode is FB

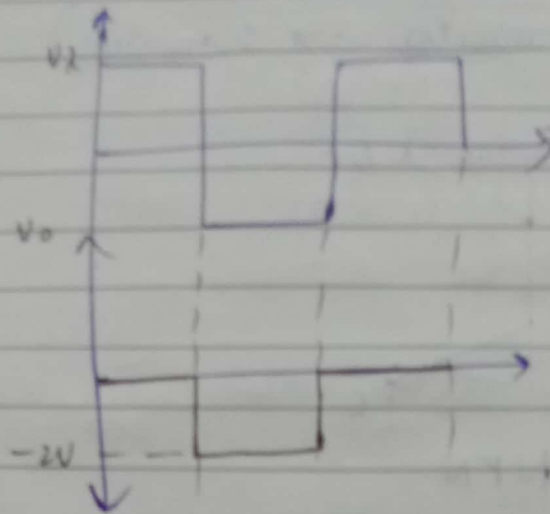
2.



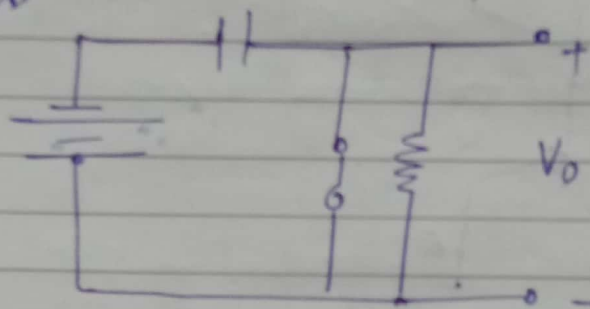
when diode is RB

$$V_o = -V - V$$

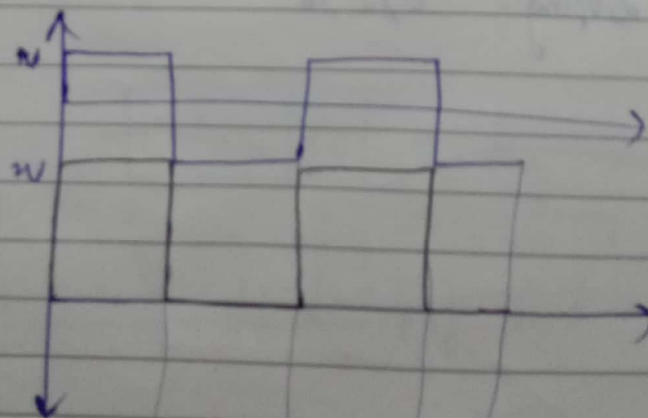
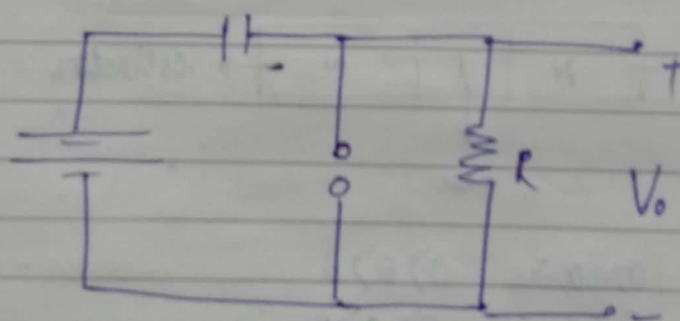
$$= -2V$$



Diode is FB



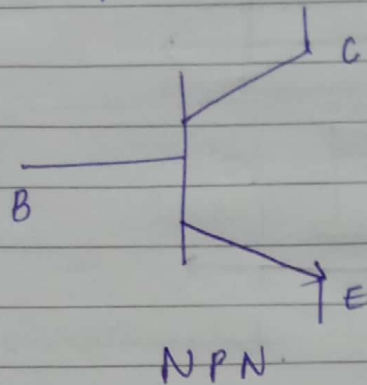
Diode is KB



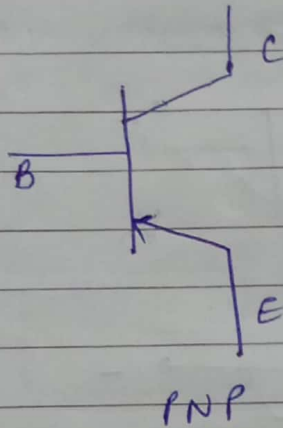
Rectifier will be done in LAB:

Page No.			
Date			

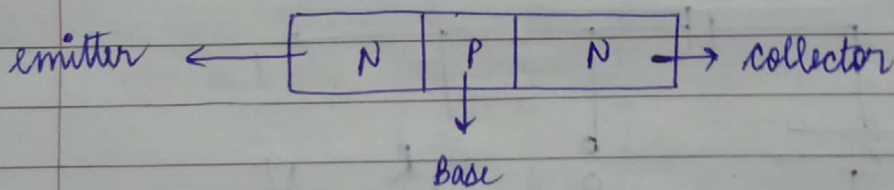
TRANSISTOR
(Transfer of resistor $\rightarrow$ transistor)



NPN

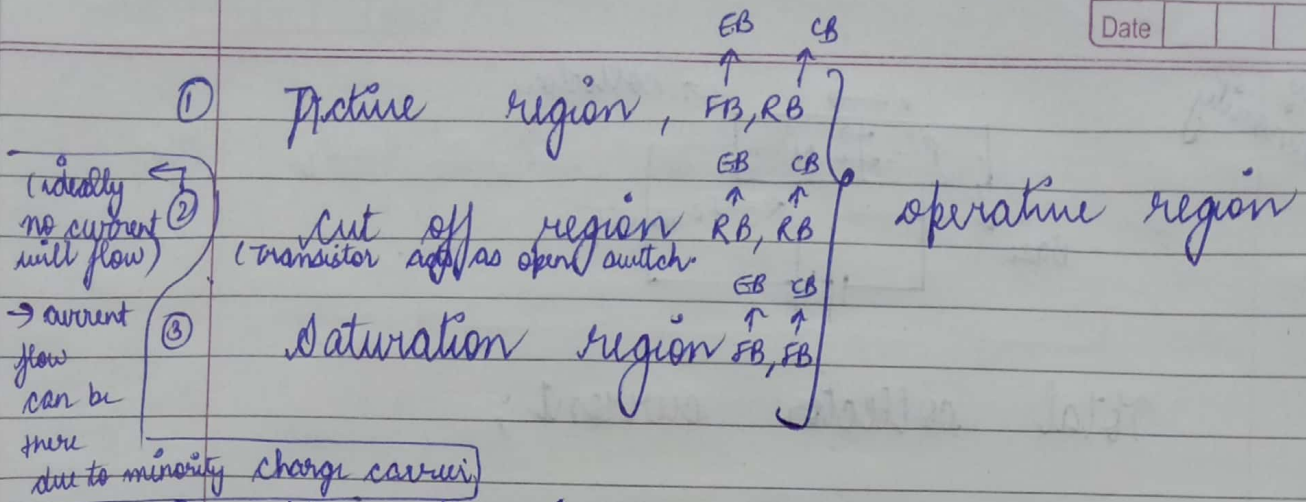


PNP



area: $C > E > B$

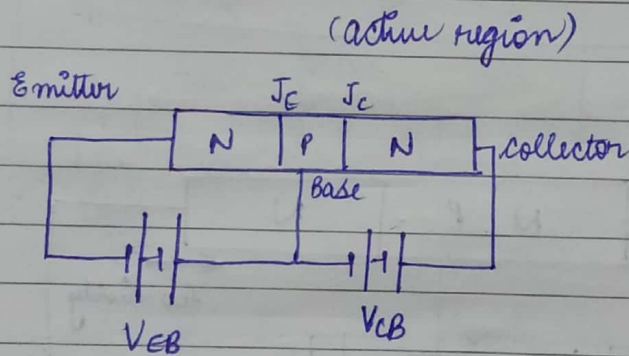
doping: $E > C > B$



How transistor acts as an amplifier.

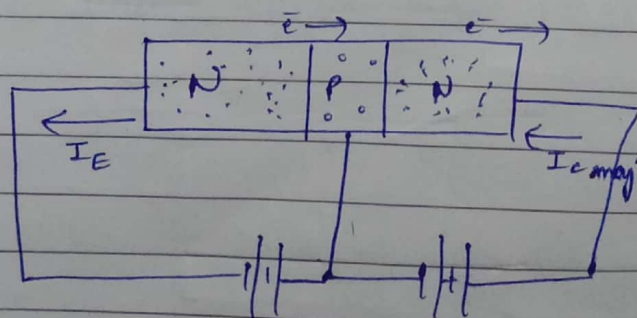
- only in active region of amplifier
- in cut off region & saturation region of it acts as switch

Biasing



In active region :

- EB junction should be forward biased
- CB junction should be reverse biased

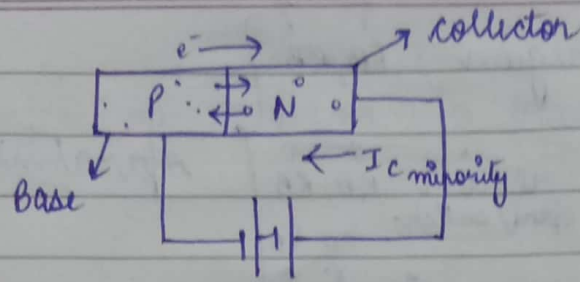


99% current is I_C , I_B is very negligible

$I_E = I_B + I_C$

↑ ↑
minority majority

now
in minority



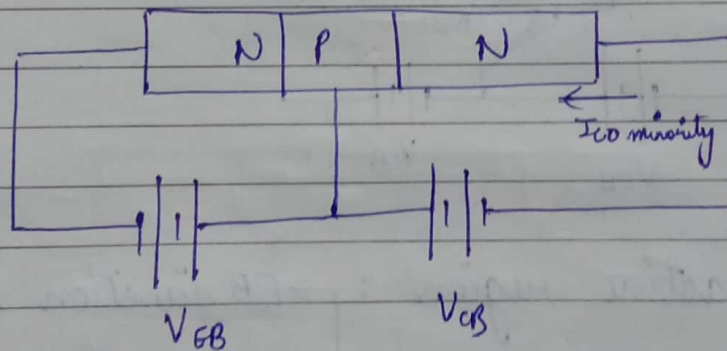
Total collector current;

$$I_c = (I_{c \text{ minority}}) + (I_{c \text{ majority}})$$

(Leakage current is the I_{c0} current due to minority).
↳ it is very very less.

③

Saturation



→ No depletion layer will form.

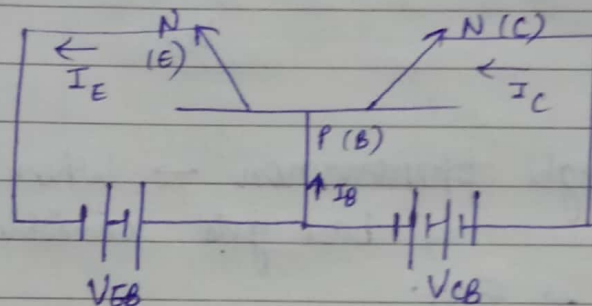
Configuration of transistor :

1) CB
(common base)

(only this is in syllabus)
2) CE
(common emitter)

3) CC
(common collector)

① Common Base (CB) configuration :



(arrows show direction of current)

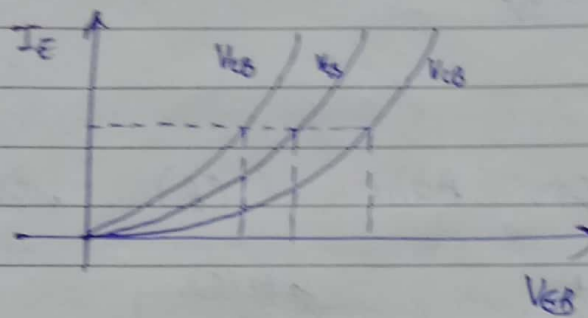
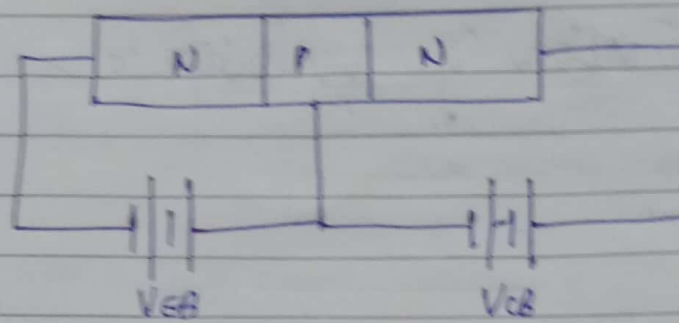
α is current gain in common configuration.

$$\alpha = \frac{I_C}{I_E} \quad (\text{if } I_C \approx I_E, \alpha \rightarrow 1)$$

α should be 1 (max^m).

BTW α value varies from (0.98-0.99)

input characteristics

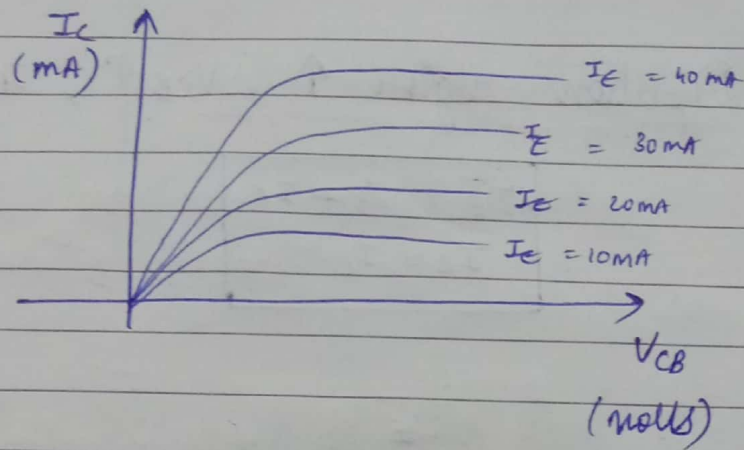


Punch through phenomenon $\rightarrow$ When due to V_{CE} base gets invisible, i.e. base is over.

Base width modulation

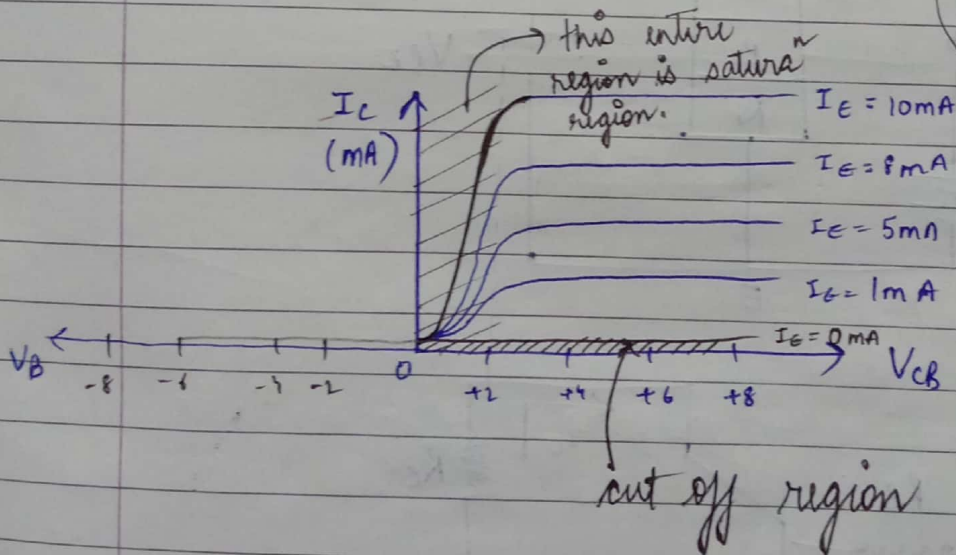
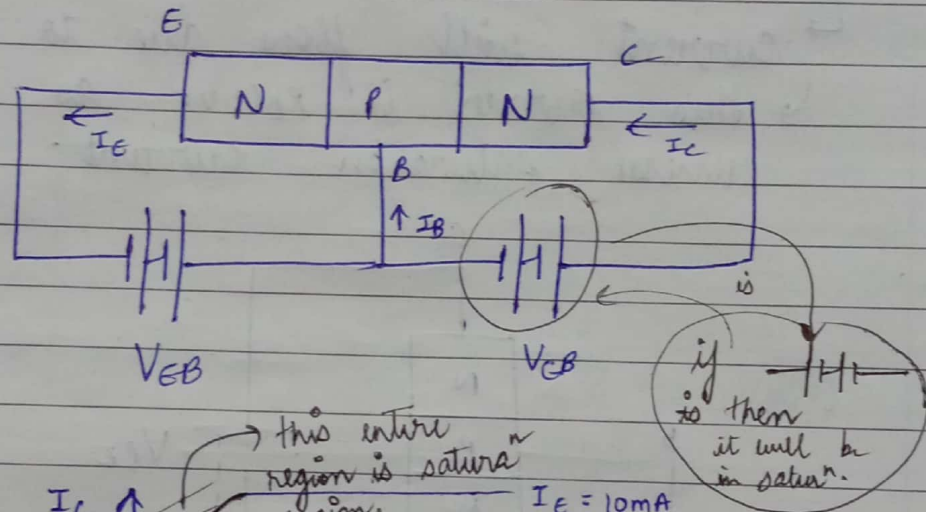
If $V_{CE} \uparrow$, $I_B \downarrow$ & $I_C \uparrow$

output characteristics



Common Base (CB) output characteristics:

(it is in active region)



Active region

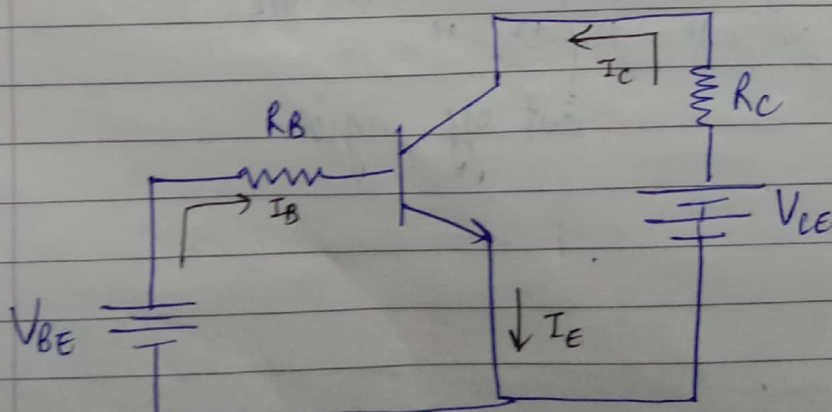
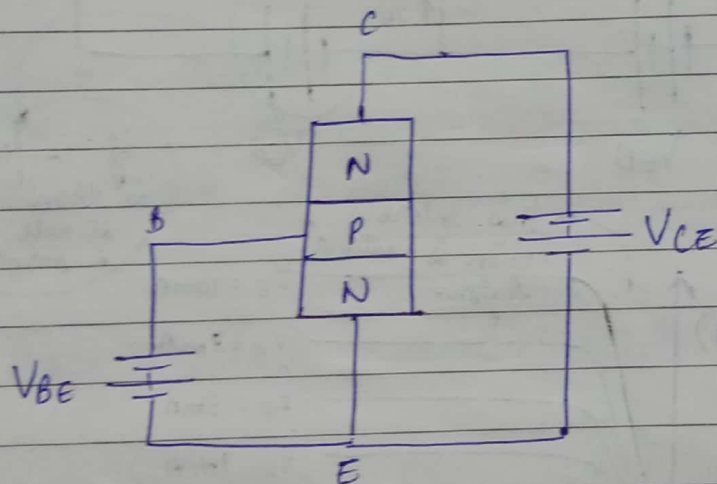
if Depletion region $\uparrow$, $V_{CB} \uparrow$, $I_B \downarrow$, $I_C \uparrow$

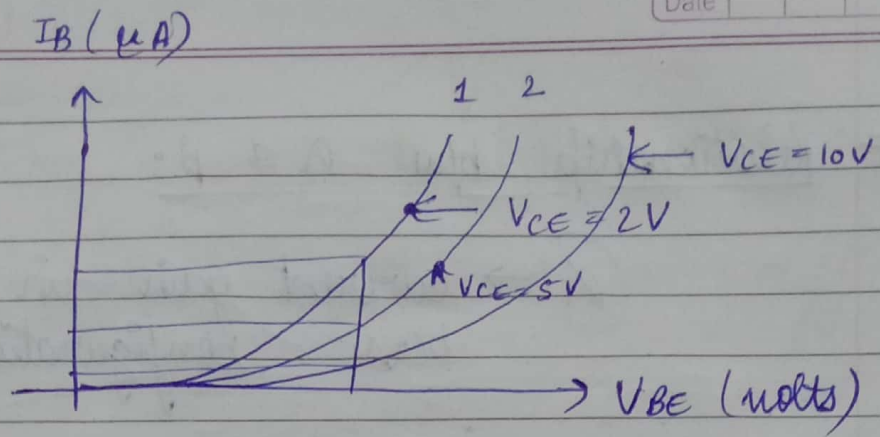
$$I_E = I_B + I_C$$

$$I_E \approx I_C$$

When both V_{EB} & V_{CB} are in cut off region

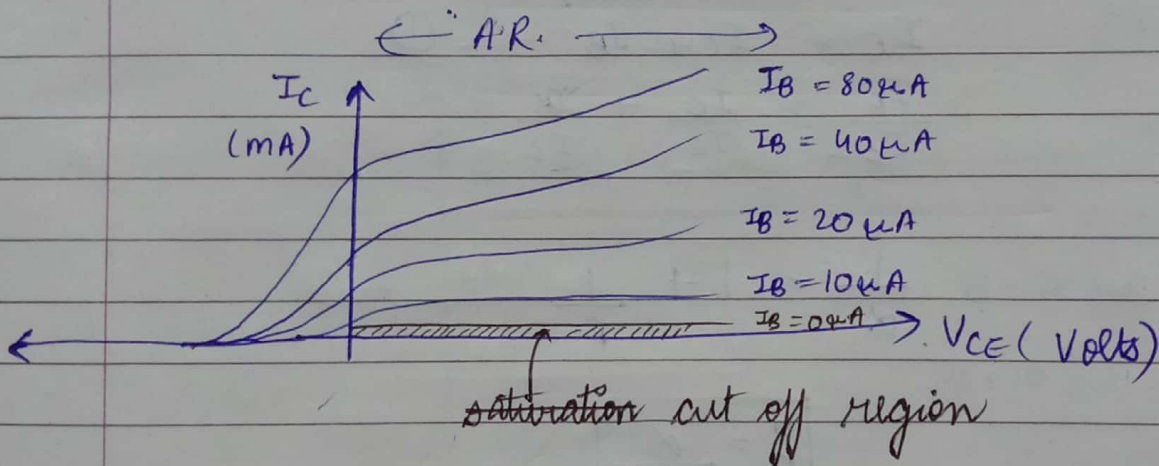
$\hookrightarrow$ current will flow due to minorities.
 $\hookrightarrow$ this current is known as reverse saturation current





1 (Keep $V_{CE} \rightarrow$ const. for one particular value)

2 (now V_{CE} is increased)



$$\beta = \frac{I_C}{I_B}$$

100 to 250

common emitter (amplifier)

→ Input impedance low
→ Output impedance high

CB emitter circuit

→ stability is very high

Relationship b/w α & β :

α → current gain in common base configuration

$$\alpha = \frac{I_C}{I_E}$$

cb configuration

(max^m value for α is 0.99)

(β is highly temp^r dependent)

$$\beta = \frac{I_C}{I_B}$$

ce confⁿ

(max^m value of β 100-250)

$$I_E = I_C + I_B \quad \text{--- ①}$$

$$\frac{I_E}{I_C} = \frac{I_C}{I_C} + \frac{I_B}{I_C}$$

$$\frac{1}{\alpha} = 1 + \frac{1}{\beta}$$

$$\boxed{\beta = \frac{\alpha}{1-\alpha}}$$

stability

$$s = 1 + \beta \quad \left. \vphantom{s = 1 + \beta} \right\} \text{highly unstable}$$

for cb, $s = 1$

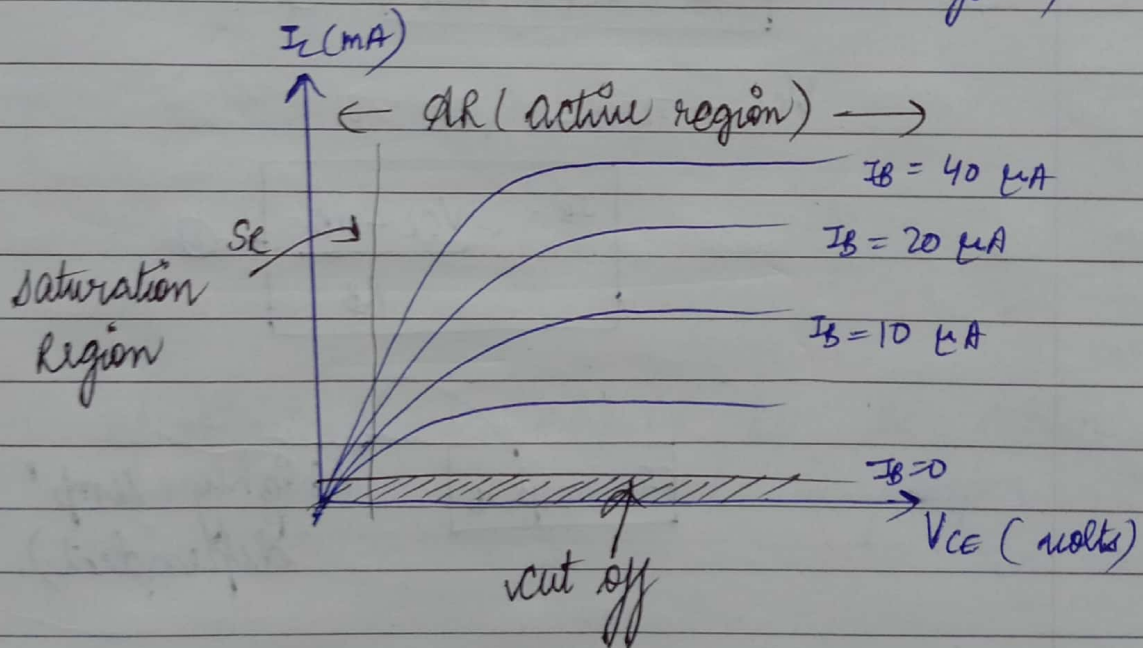
common emitter is highly unstable

to make it stable we use diff. biasing techniques.

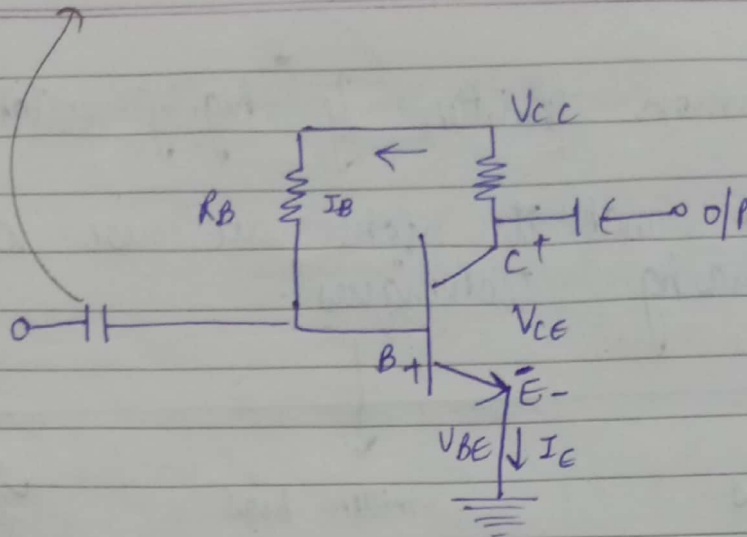
Fixed bias

Emitter bias

Voltage divider bias
(this gives most stable perm)



capacitor blocks DC passes AC.



KVL

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$\boxed{V_{CE} = V_{CC} - I_C R_C} \quad \text{--- (1)}$$

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$\boxed{I_B = \frac{V_{CC} - V_{BE}}{R_B}} \quad \text{--- (2)}$$

$$\boxed{I_C = \beta I_B} \quad \text{(Highly temp}^r \text{ dependent)}$$

Load line:

→ if we want a point on x axis then its y coordinate should be zero.

in eqⁿ --- (one) of AT graph

where, $I_C = 0$

$$V_{CE} = V_{CC} - 0 \times R_C$$

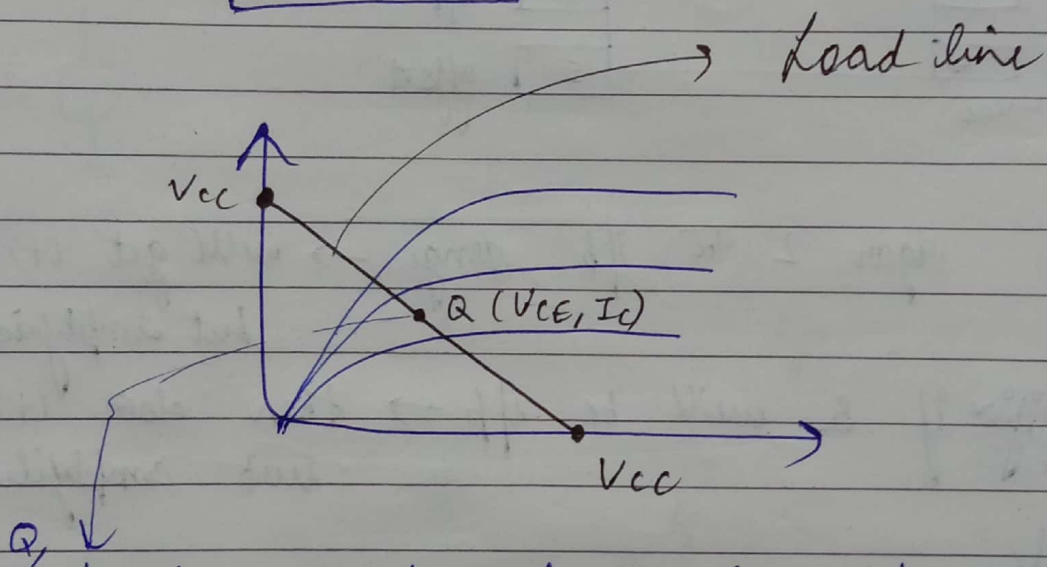
$$\boxed{V_{CE} = V_{CC}}$$

when we want a point on y axis, then x coordinate should be zero.

$$eq^n - (1)$$

$$0 = V_{CC} - I_C R_C$$

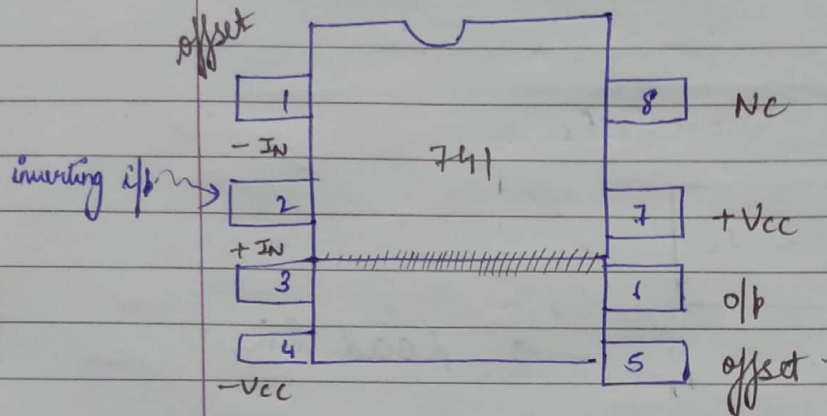
$$\boxed{\frac{V_{CC}}{R_C} = I_C}$$



operating point should be at center so that the output is proper.

Operational amplifier (op-amp) - 741 IC

Ideal characteristics



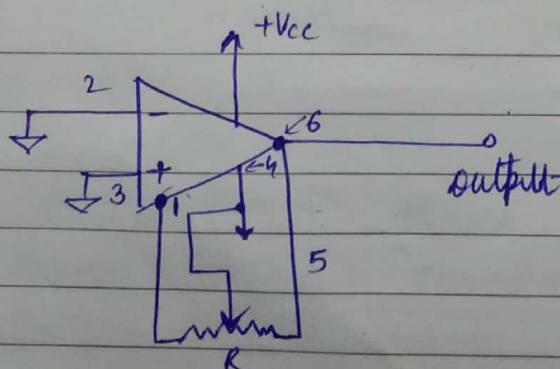
① Inverting amplifier

Agar 2 ka i/p denge $\rightarrow$ we'll get (+)ve o/p but amplified - ①

② Non inverting amplifier

if 3, we'll be i/p $\rightarrow$ then also (+)ve o/p but amplified - ②

③ Differential



if 2 & 3 are grounded then also we get some output in multimeter, so as to get output as equal to zero, set the value of R accordingly

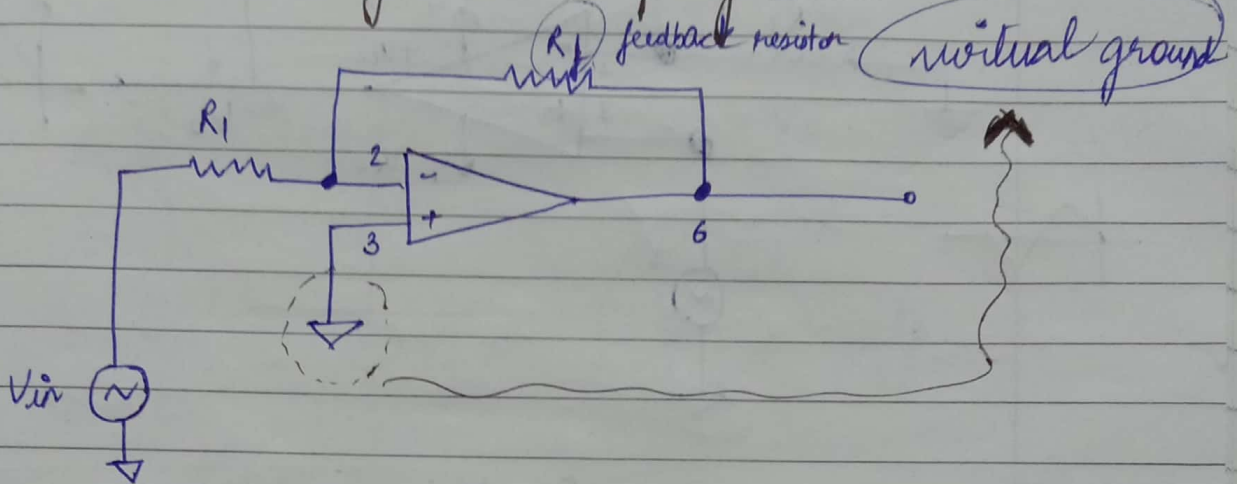
so as to get the o/p as 0

A

Inverting amplifier

eg i/p $\uparrow$
o/p $\downarrow$

(-)ve feedback



$$\frac{V_{in} - 0}{R_1} = \frac{0 - V_o}{R_f}$$

$$\frac{V_{in}}{R_1} = \frac{-V_o}{R_f}$$

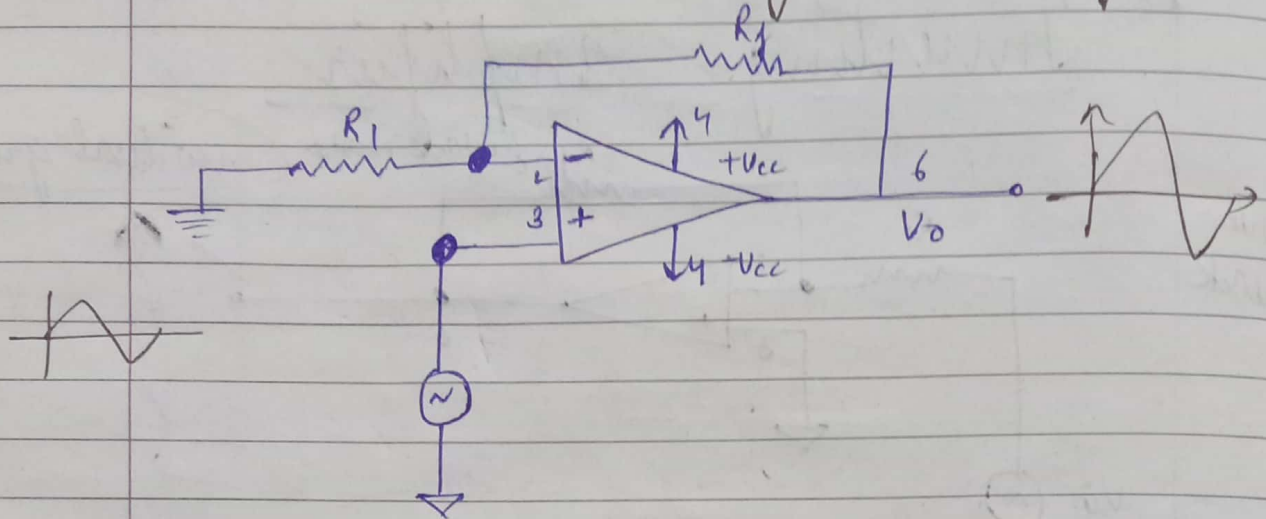
$$\boxed{\frac{V_o}{V_{in}} = \frac{-R_f}{R_1}}$$

$$\boxed{V_o = -\frac{R_f}{R_1} V_{in}}$$

P4

2 → inverting
3 → non-inverting

* Non Inverting Amplifier :



$$\frac{0 - V_{in}}{R_1} = \frac{V_{in} - V_o}{R_f}$$

$$\frac{V_o}{V_{in}} =$$

$$\frac{-V_{in}}{R_1} = \frac{V_{in} - V_o}{R_f}$$

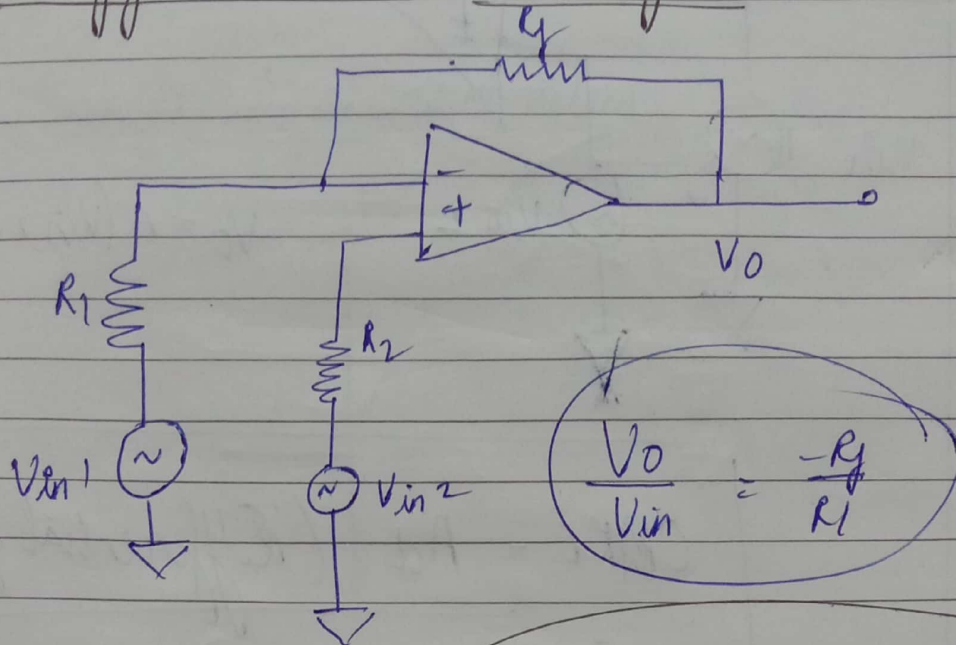
$$\frac{V_o}{R_f} = \frac{V_{in}}{R_f} + \frac{V_{in}}{R_1}$$

gain → $\boxed{\frac{V_o}{V_{in}} = R_f \left(\frac{1}{R_f} + \frac{1}{R_1} \right)}$

$$V_o = \left(1 + \frac{R_f}{R_1} \right) V_{in}$$

(not that much imp)

Differential amplifier:

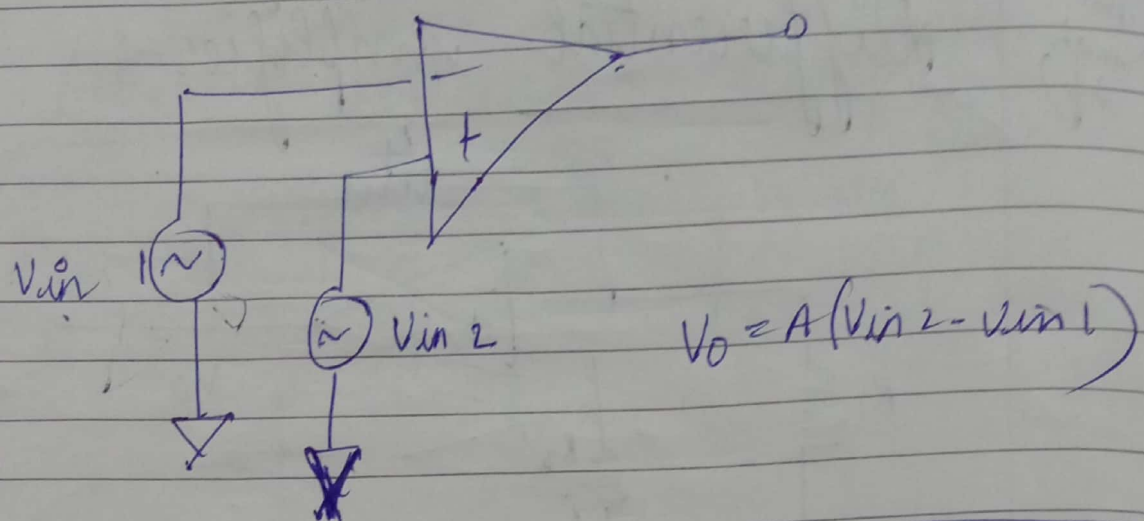


$$\frac{V_o}{V_{in}} = \frac{-R_f}{R_1}$$

$$V_o = \frac{-R_f}{R_1} (V_{in2} - V_{in1})$$

Ideal characteristics of op-amp →

- 1) ∞ i/p ~~imp~~ impedance
- 2) zero o/p impedance
- 3) gain $\rightarrow \infty$
- 4) CMRR $\rightarrow \infty$
- 5) ~~slow~~ slew rate $\rightarrow \infty$ common mode rejection ratio
- 6) Bandwidth $\rightarrow \infty$



$$CMRR = \frac{A_d \text{ (Differential gain)}}{A_{CM} \text{ (common mode)}}$$

$$CMRR = \frac{A_d}{0} = \infty$$

Slew rate

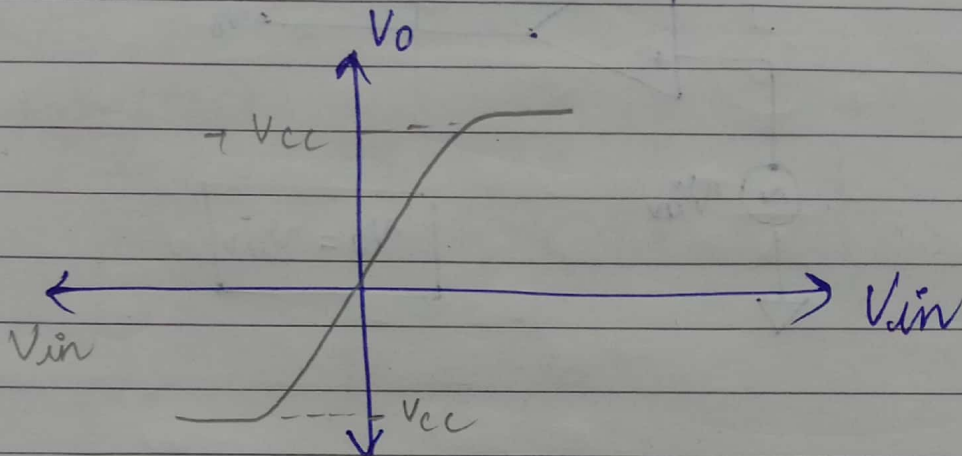
$$SR = \frac{dV_o}{dt}$$

(input kitna fast o/p k jaldi respond
hoga uski)

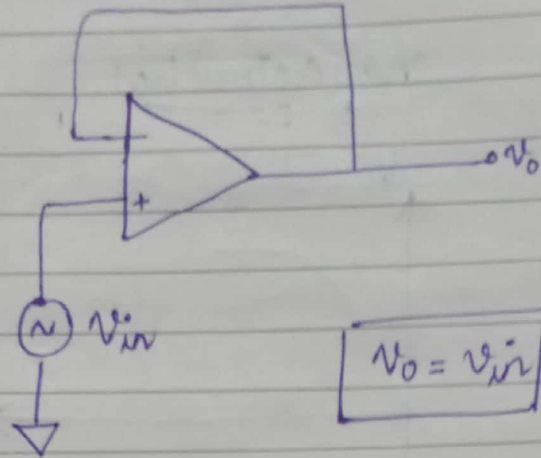
Bandwidth

↳ range of ν .

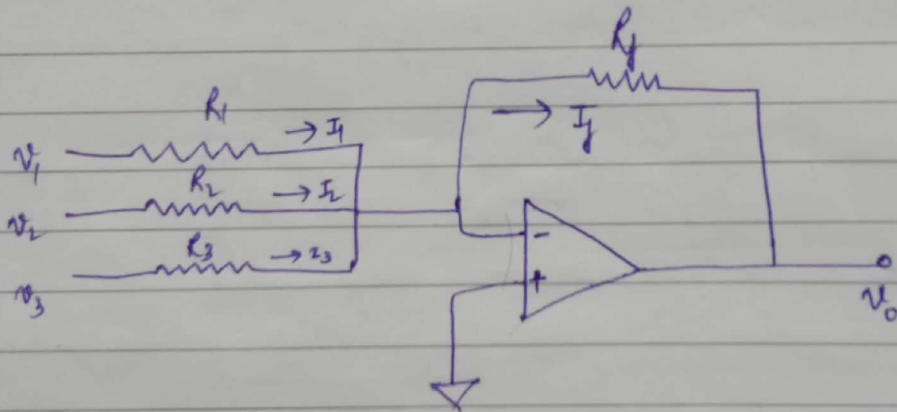
Transfer characteristics



① Unity followers



② Summation $\rightarrow$ or adder circuit



$$I_1 + I_2 + I_3 = I_f$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_o}{R_f}$$

$$V_o = -R_f \left[\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right]$$

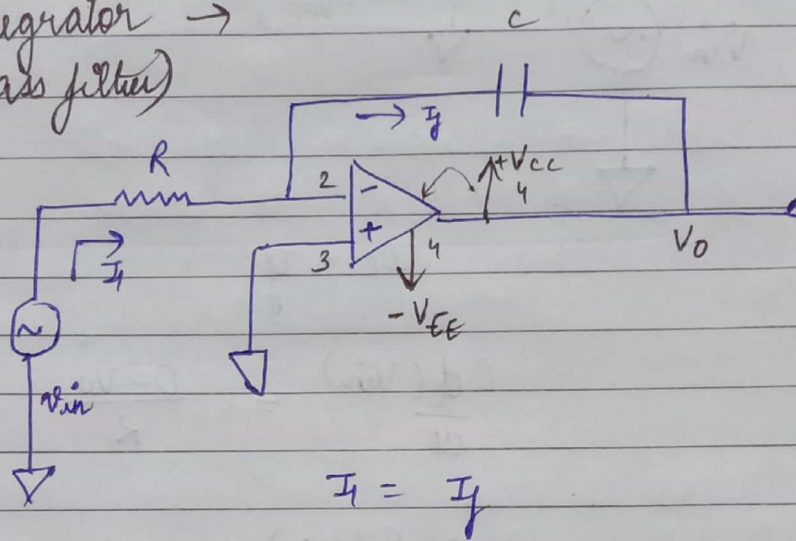
$$\therefore V_o = -\frac{R_f}{R} [V_1 + V_2 + V_3]$$

$$R_f = R$$

$$V_o = -(V_1 + V_2 + V_3)$$

→ blocks DC passes AC

③ Integrator
(low pass filter)



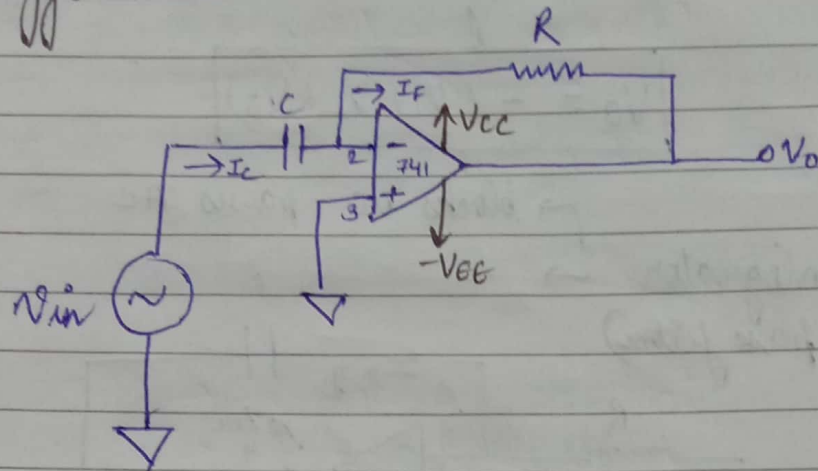
$$\frac{V_{in} - 0}{R} = C \frac{d}{dt} [0 - V_o]$$

$$\frac{V_{in}}{R} = -C \frac{dV_o}{dt}$$

$$\frac{-1}{RC} V_{in} = \frac{dV_o}{dt}$$

$$V_o = \frac{-1}{RC} \int V_{in} dt$$

4) Differentiator :



$$I_c = I_f$$

$$C \frac{d(v_{in})}{dt} = \frac{0 - v_o}{R}$$

$$C \frac{d(v_{in})}{dt} = \frac{-v_o}{R}$$

$$-RC \frac{d(v_{in})}{dt} = v_o$$

$$v_o = -RC \frac{d v_{in}}{dt}$$

Digital Electronics :

- ① Boolean algebra
- ② Logic gates
- ③ K-map upto 4 variables
- ④ Full addition and subtraction } combination ckts.

Boolean algebra:

- 1) $A \cdot \bar{A} = 0$
- 2) $A + \bar{A} = 1$
- 3) $(A+B)+C = A+(B+C)$
- 4) $(A \cdot B) \cdot C = A \cdot (B \cdot C)$
- 5) De-morgan's theorem
- 6) $\overline{A \cdot B} = \bar{A} + \bar{B}$
- 7) $\overline{(\bar{A} + \bar{B})} = A \cdot B$

Q
Solⁿ

$$F = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + A\bar{C}\bar{B}$$

$$F = \bar{A}\bar{B}(C+\bar{C}) + A(B\bar{C} + \bar{B}C)$$

$$F = \bar{A}\bar{B}(1) + A(B\bar{C} + \bar{B}C)$$

$$F = \bar{A}\bar{B}(1) + A(B\bar{C} + \bar{B}C) \quad \text{Ans.}$$

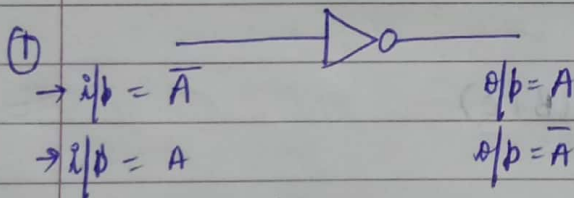
H

SOP $\rightarrow$ Sum of product

POS $\rightarrow$ Product of sum

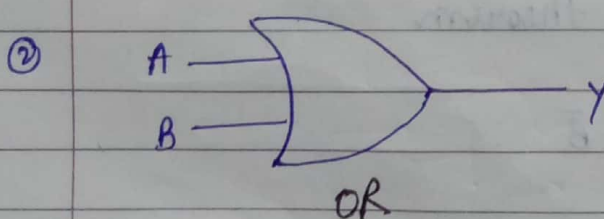
$\rightarrow$ eg: $(\bar{A}+\bar{B}+\bar{C})(A+B+C)(\bar{A}+B+C)$

Logic gate :

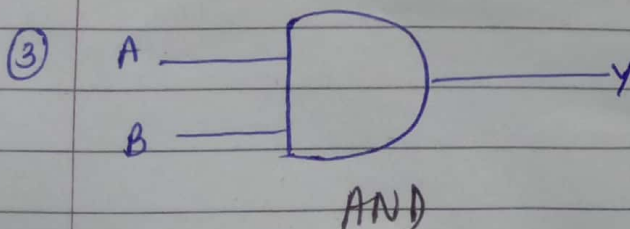


NOT

i/p	o/p
0	1
1	0

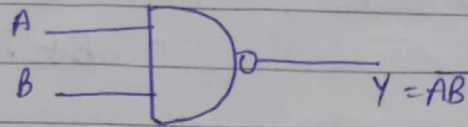


A	B	o/p
1	1	1
0	0	0
1	0	1
0	1	1



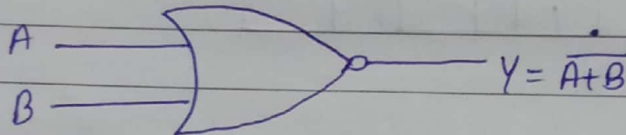
A	B	o/p
0	0	0
0	1	0
1	0	0
1	1	1

④ NAND gate :



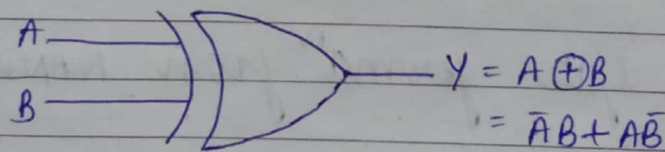
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

⑤ NOR gate :



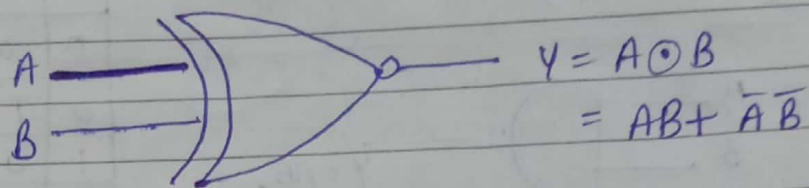
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

⑥ EX-OR gate
↓
Exclusive



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

⑦ Ex - NOR gate

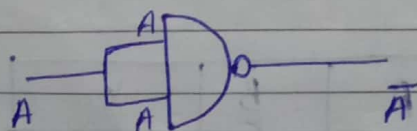


A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

NAND & NOR are universal gates
($\because$ we can make any basic gate (OR, NOT, AND))

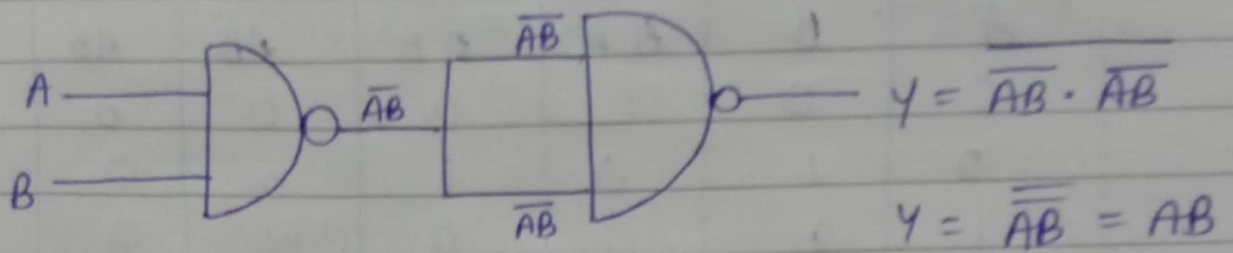
Universal gate NAND :

① NOT gate formation from NAND

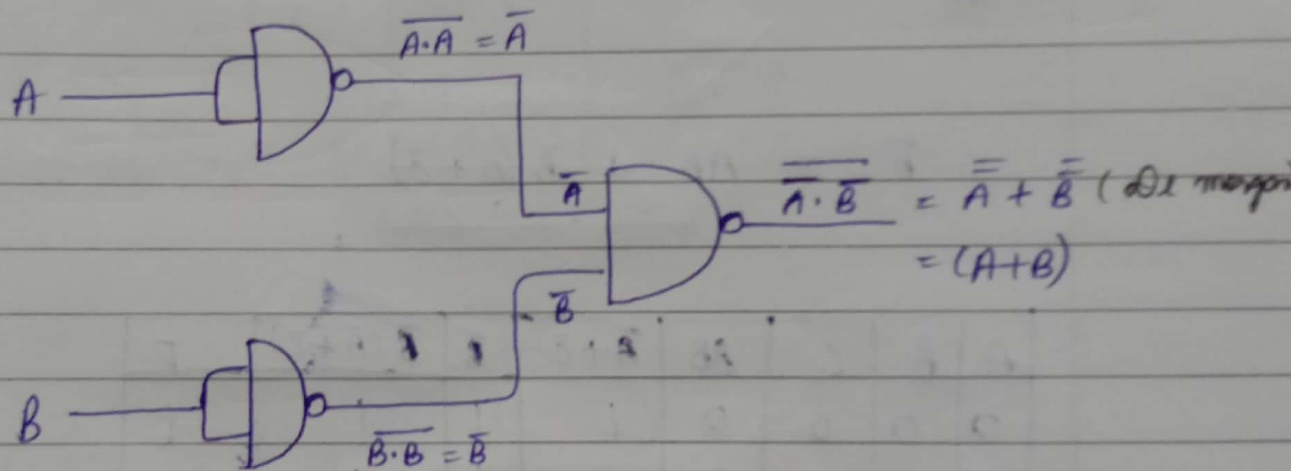


$$Y = \overline{A \cdot A} = \bar{A}$$

② AND gate :



③ OR gate :



Universal gate

Minterm denoted by Σm .

Minterms & sum of product :

		M_0	M_1	M_2	M_3
A	B	$\bar{A}\bar{B}$	$\bar{A}B$	$A\bar{B}$	AB
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

$$\Sigma m = \bar{A}\bar{B}, \bar{A}B, A\bar{B}, AB$$

$$F = AB + \bar{B}(\bar{A} + \bar{C})$$

A	B	C	AB	$\bar{A} + \bar{C}$	$\bar{B}$	$\bar{B}(\bar{A} + \bar{C})$	F	
0	0	0	0	1	1	1	1	0
0	0	1	0	1	1	1	1	1
0	1	0	0	1	0	0	0	2
0	1	1	0	1	0	0	0	3
1	0	0	0	1	1	1	1	4
1	0	1	0	0	1	0	0	5
1	1	0	1	1	0	0	1	6
1	1	1	1	0	0	0	1	7

SOP form of $F \rightarrow \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} + ABC$

$$\Sigma m (0, 1, 4, 6, 7)$$

POS form:

$$F = \pi_m (2, 3, 5)$$

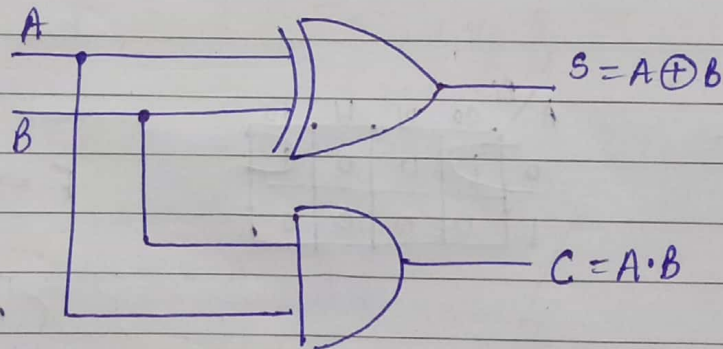
$$F = (A + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + B + \bar{C})$$

half adder

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

OK

$$C = A \cdot B$$



K-map

2 variable K map:

A \ B	0	1
0	0	1
1	1	0

Sum

$$\begin{aligned} \text{Sum} &= \bar{A}B + A\bar{B} \\ &= A \oplus B \end{aligned}$$

for carry

A \ B	0	1
0	0	0
1	0	1

$$C = \text{carry} = AB$$

3 variable K-maps-

A \ BC	00	01	11	10
0	0	0	1	1
1	1	1	0	0

$$F = AB + \bar{A}B$$

by folding

A \ BC	00	01	11	10
0	1	0	0	1
1	0	0	0	0

$$F = \bar{A}\bar{C}$$

A \ BC	00	01	11	10
0	1	1	1	1
1	1	1	1	1

Try to group
max^m no.
of elements together

Q
same
question

$$F = \bar{x}yz + \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}\bar{y}z + x\bar{y}\bar{z} + x\bar{y}z + \bar{x} + y\bar{z}$$

OR

$$F = \sum m(0, 1, 3, 4, 5, 6)$$

Simplify the expression using K map:-

		yz			
		00	01	11	10
x	0	1	1	1	1
	1	1			1

$$F = \bar{x} + \bar{y}\bar{z} + y\bar{z}$$

$$F = \bar{x} + \bar{z}(y + \bar{y})$$

$$F = \bar{x} + \bar{z}$$

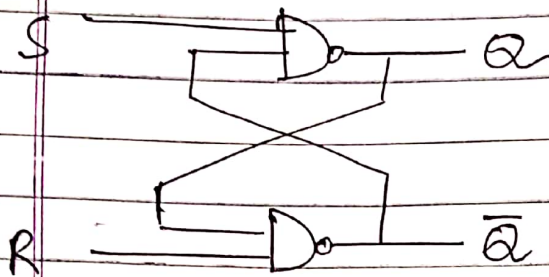
Q $F(C, A, B) = \pi(1, 2, 3, 7)$

		AB			
		00	01	11	10
C	0	0	1	1	0
	1	0	0	0	0

POS form: $F = (C + \bar{B})(\bar{A} + B)(C + \bar{A})$

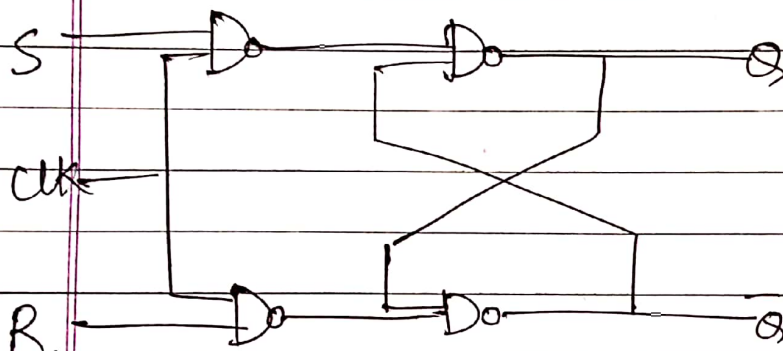
Thurs

SR LATCH



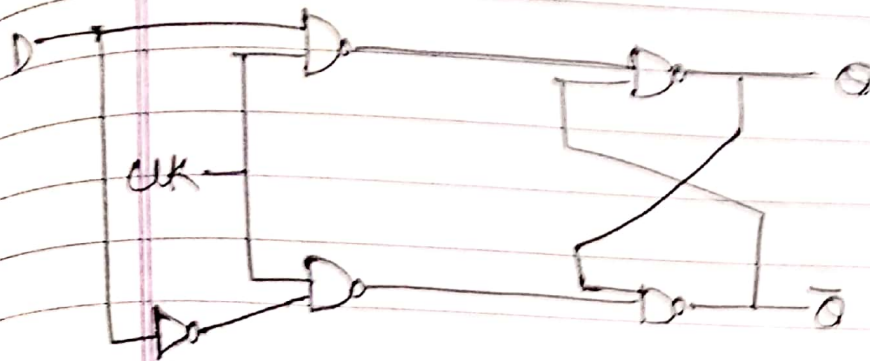
	S	R	Q	Q̄
	1	0	0	1
	1	1	0	1
<u>NAND</u>	0	0	1	0
	0	1	1	0
	1	0	1	1
	1	1	0	1

→ Intermediate



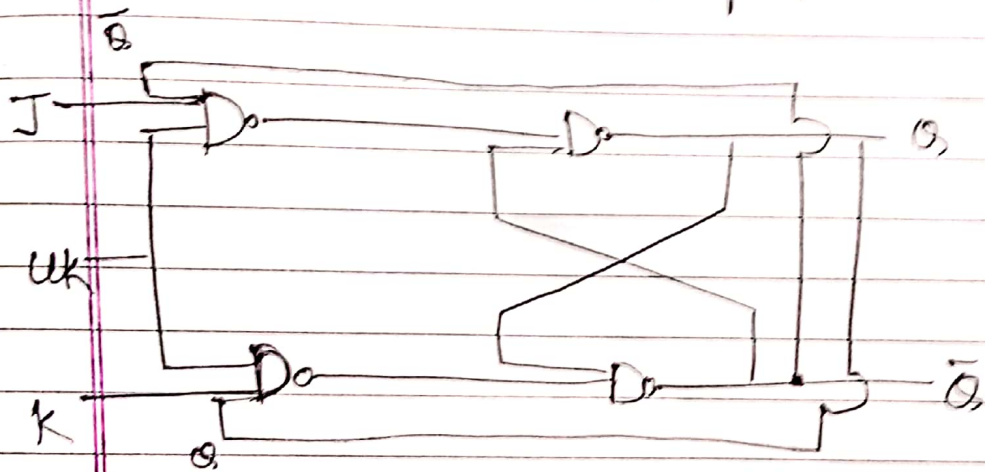
C	S	R	Next state of Q
0	x	x	NC
1	0	0	NC
1	0	1	Q=0, Reset state
1	1	0	Q=1, Set state
1	1	1	Intermediate

D-latch (flip-flop)



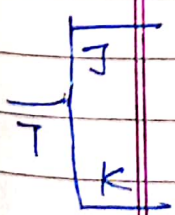
C	D	Next state of Q
0	X	NC
1	0	0
1	1	1

JK flip flop:



J	K	Q	Q-bar
0	0	No change	
0	1	0	1
1	0	1	0
1	1	Q.n	

T-flip flop:



C	T	Next state of Q
0	X	NC
1	0	1
1	1	0